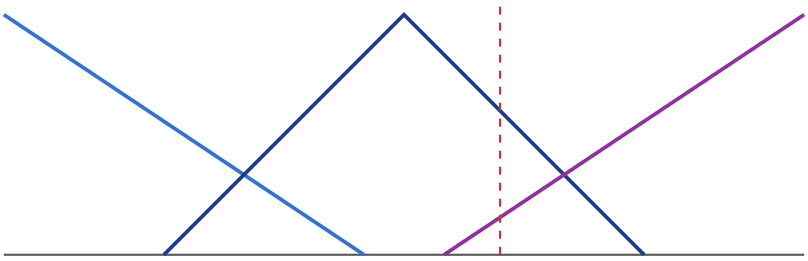
MA4144
Neural Networks and Fuzzy Logic

Study Notes - Part 2: Fuzzy Logic



University of Moratuwa
Semester 7

Fuzzy lessons 1–7

Classical and fuzzy sets · Relations and composition · Membership functions and defuzzification
Classical and fuzzy logic · Linguistic variables and hedges · Mamdani, Sugeno and Tsukamoto inference

From crisp sets to fuzzy inference systems, with every operator and defuzzification method worked through.

1 CLASSICAL SETS AND FUZZY SETS

- The universe of discourse is the universe of all available information on a given problem.
- Once this universe is defined, we are able to define certain events on this information space.
- Following Figure shows an abstraction of a universe of discourse, say X, and a crisp (classical) set A somewhere in this universe.
- A classical set is defined by crisp boundaries, that is, there is no uncertainty in the prescription or location of the boundaries of the set, as shown in Figure below where the boundary of crisp set A is an unambiguous line.
- A fuzzy set, on the other hand, is prescribed by vague or ambiguous properties; hence, its boundaries are ambiguously specified, as shown by the Fuzzy boundary for set A.

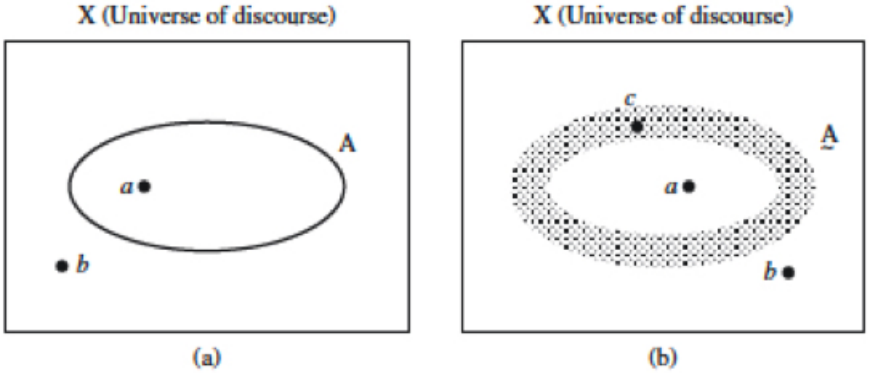


Figure 1.1: (a) crisp set boundary and (b) fuzzy set boundary

1.1 CLASSICAL SETS

- Define a universe of discourse, X, as a collection of objects all having the same characteristics.
- The individual elements in the universe X will be denoted as x.
- The features of the elements in X can be discrete, countable integers, or continuous valued quantities on the real line. Examples:
  - the clock speeds of computer CPUs;
  - the operating currents of an electronic motor;
  - the operating temperature of a heat pump (in degrees Celsius);
  - the Richter magnitudes of an earthquake; the integers 1 to 10.
- Collections of elements within a universe are called **sets**, and collections of elements within sets are called **subsets**. The collection of all possible sets in the universe is called the **whole set**.
- For crisp sets A and B consisting of collections of some elements in X, the following notation is defined:

$x \in X \rightarrow x$ belongs to X
$x \in A \rightarrow x$ belongs to A
$x \notin A \rightarrow x$ does not belong to A

- For sets A and B on X, we also have

$A \subset B \rightarrow A$ is fully contained in B (if $x \in A$, then $x \in B$)
$A \subseteq B \rightarrow A$ is contained in or is equivalent to B
$A \leftrightarrow B \rightarrow A \subseteq B$ and $B \subseteq A$ (A is equivalent to B)

- The null set, $\emptyset$, as the set containing no elements, and the whole set, X, as the set of all elements in the universe.
- All possible sets of X constitute a special set called the power set, $P(X)$.
- The total number of elements in a universe X is called its cardinal number; denoted n_x, where x again is a label for individual elements in the universe.

1.2 Operations on Classical or Crisp Sets

- The union $A \cup B$ of crisp sets A and B is the crisp set of elements that are members of A or B or both.
  - In general, the union $B_1 \cup B_2 \cdots \cup B_n$ of $n \geq 1$ crisp sets $B_1, B_2, \ldots, B_n$ is

$$\cup_{i=1}^{n} B_i = \{x | x \in B_i \text{ for at least one value of } i \text{ from } 1 \text{ to } n\}$$

- The intersection $A \cap B$ of crisp sets A and B is the crisp set of elements that are members of both A and B.
  - In general, the intersection $B_1 \cap B_2 \cdots \cap B_n$ of $n \geq 1$ crisp sets $B_1, B_2, \ldots, B_n$ is

$$\cap_{i=1}^{n} B_i = \{x | x \in B_i \text{ for all values } i \text{ from } 1 \text{ to } n\}$$

- The complement $\overline{B}$ of a crisp set B is a crisp set containing those elements of the universal set X that are not members of B.
- The difference $A|B$ between crisp sets A and B is the set of those elements that belong to A, but do not belong to B.

1.3 Properties of Classical (Crisp) Sets

- The following properties hold for universal set X, empty set $\emptyset$, and any crisp sets A, B, and C:

Table 1.1

Commutativity	$A \cup B = B \cup A$
	$A \cap B = B \cap A.$
Associativity	$A \cup (B \cup C) = (A \cup B) \cup C$
	$A \cap (B \cap C) = (A \cap B) \cap C.$
Distributivity	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$
Idempotency	$A \cup A = A$
	$A \cap A = A.$
Identity	$A \cup \emptyset = A$
	$A \cap X = A$
	$A \cap \emptyset = \emptyset.$
	$A \cup X = X.$
Transitivity	If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C.$
Involution	$\overline{\overline{A}} = A.$

- Two special properties of set operations are known as the excluded middle axioms and De Morgan's principles.

Axiom of the excluded middle	$A \cup \overline{A} = X$
Axiom of the excluded contrdiction	$A \cap \overline{A} = \emptyset$

- De Morgan's principles are important because of their usefulness in proving tautologies and contradictions in logic, as well as in a host of other set operations and proofs.

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$
$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

- In general, De Morgan's principles can be stated for n sets, as provided here for events, E_i :

$$\overline{E_1 \cap E_2 \cap \cdots \cap E_n} = \overline{E_1} \cup \overline{E_2} \cup \cdots \cup \overline{E_n}$$
$$\overline{E_1 \cup E_2 \cup \cdots \cup E_n} = \overline{E_1} \cap \overline{E_2} \cap \cdots \cap \overline{E_n}$$

Example 1. A shallow arch consists of two slender members as shown in Figure below If either of the members fails, then the arch will collapse.

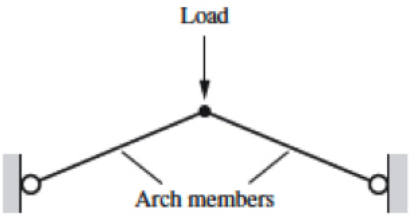


Figure 1.2

If E_1 = survival of member 1 and E_2 = survival of member 2, then survival of the arch = $E_1 \cap E_2$, and, conversely, collapse of the arch = $\overline{E_1 \cap E_2}$. Logically, collapse of the arch will occur if either of the members fails, that is, when $\overline{E_1} \cup \overline{E_2}$. Therefore, $\overline{E_1 \cap E_2} = \overline{E_1} \cup \overline{E_2}$, which is an illustration of De Morgan's principle

1.4 Mapping of Classical Sets to Functions

- If an element x is contained in X and corresponds to an element y contained in Y, it is generally termed a mapping from X to Y, or $f : X \to Y$. As a mapping, the characteristic (indicator) function χ_A is defined as

$$\chi_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases} \tag{1.1}$$

where χ_A expresses "membership" in set A for the element x in the universe.

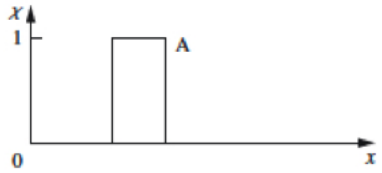


Figure 1.3

- For any set A defined on the universe X, there exists a function-theoretic set, called a value set, denoted $V(A)$, under the mapping of the characteristic function, χ. By convention, the null set $\emptyset$ is assigned the membership value 0 and the whole set X is assigned the membership value 1.
- Now, define two sets, A and B, on the universe X. The union and intersection of these two sets, the complement of a single set and containment of two sets in terms of function-theoretic terms are given as follows (the symbol $\vee$ is the maximum operator and $\wedge$ is the minimum operator):

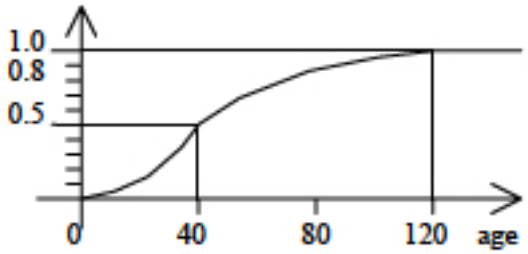
$$\text{Union} \quad A \cup B \to \chi_{A \cup B}(x) = \chi_A(x) \vee \chi_B(x) = \max\{\chi_A(x), \chi_B(x)\} \tag{1.2}$$
$$\text{Intersection} \quad A \cap B \to \chi_{A \cap B}(x) = \chi_A(x) \wedge \chi_B(x) = \min\{\chi_A(x), \chi_B(x)\} \tag{1.3}$$
$$\text{Complement} \quad \overline{A} \to \chi_{\overline{A}}(x) = 1 - \chi_A(x) \tag{1.4}$$
$$\text{Containment} \quad A \subseteq B \to \chi_A(x) \leq \chi_B(x) \tag{1.5}$$

1.5 Fuzzy Sets

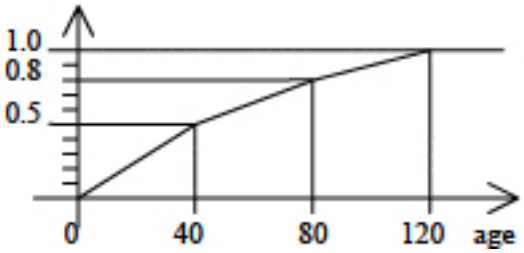
- In classical, or crisp, sets the transition for an element in the universe between membership and nonmembership in a given set is well defined.
- For an element in a universe that contains fuzzy sets, this transition can be gradual. This transition among various degrees of membership can be thought of as conforming to the fact that the boundaries of the fuzzy sets are vague and ambiguous.

Example 2. Let S be the set of all human beings, used as the universe set, and let $S_f = \{s \in S | s \text{ is old}\}$.

- Then S_f is a "fuzzy subset" of S because the property "old" is not well defined and cannot be precisely measured: given a person who is 40 year old, it is not clear if this person belongs to the set S_f.
- Thus, to make the subset S_f well defined, we have to quantify the concept "old," so as to characterize the subset S_f in a precise and rigorous way.
- One can describe the concept "old" by the curve shown in following Figures.



(a)



(b)

Figure 1.4

- In both curves :
  - The only people who are considered to be "absolutely old" are those 120 years old or older, and the only people who are considered to be "absolutely young" are those newborns.
  - Alll the other people are old as well as young, depending on their actual ages. For example, a person 40 years old is considered to be "old" with "degree 0.5" and at the same time also "young" with "degree 0.5" according to the measuring curve that we used.
  - Thus, the curves establishe a mathematical measure for the "oldness" of a human being, and hence can be used to define the partial membership of any person relative to the subset S_f described above.
- Both are reasonable and acceptable in common sense.

- A fuzzy set, denoted $\underset{\sim}{A}$, is a set containing elements that have varying degrees of membership in the set.
- Elements of a fuzzy set are mapped to a universe of membership values using a function-theoretic form. This function maps elements of a fuzzy set $\underset{\sim}{A}$ to a real numbered value on the interval $[0, 1]$.
- If an element in the universe, say x, is a member of fuzzy set $\underset{\sim}{A}$, then this mapping is given by $\mu_{\underset{\sim}{A}}(x) \in [0, 1]$.

- A notation convention for fuzzy sets when the universe of discourse, X, is discrete and finite, is as follows for a fuzzy set $\underset{\sim}{A}$:

$$\underset{\sim}{A} = \left\{ \frac{\mu_{\underset{\sim}{A}}(x_1)}{x_1} + \frac{\mu_{\underset{\sim}{A}}(x_2)}{x_2} + \dots \right\} = \left\{ \sum_i \frac{\mu_{\underset{\sim}{A}}(x_i)}{x_i} \right\} \tag{1.6}$$

- When the universe, X, is continuous and infinite, the fuzzy set $\underset{\sim}{A}$ is denoted as

$$\left\{ \int \frac{\mu_{\underset{\sim}{A}}(x)}{x} \right\} \tag{1.7}$$

NOTE:

- In both notations, the horizontal bar is not a quotient but rather a delimiter.
- The numerator in each term is the membership value in set $\underset{\sim}{A}$ associated with the element of the universe indicated in the denominator.
- In the first notation, the summation symbol is not for algebraic summation, but rather denotes the collection or aggregation of each element; hence, the "+" signs in the first notation are not the algebraic "add" but are an aggregation or collection operator.
- In the second notation, the integral sign is not an algebraic integral but a continuous function-theoretic aggregation operator for continuous variables.

1.5.1 Fuzzy Set Operations

Define three fuzzy sets $\underset{\sim}{A}, \underset{\sim}{B}$, and $\underset{\sim}{C}$ on the universe X.

- Operations of union, intersection, and complement are defined for $\underset{\sim}{A}, \underset{\sim}{B}$, and $\underset{\sim}{C}$ on the universe X is defined as follows:

$$\text{Union} \quad \mu_{\underset{\sim}{A}\cup\underset{\sim}{B}}(x) = \mu_{\underset{\sim}{A}}(x) \vee \mu_{\underset{\sim}{B}}(x) \tag{1.8}$$

$$\text{Intersection} \quad \mu_{\underset{\sim}{A}\cap\underset{\sim}{B}}(x) = \mu_{\underset{\sim}{A}}(x) \wedge \mu_{\underset{\sim}{B}}(x) \tag{1.9}$$

$$\text{Complement} \quad \mu_{\overline{\underset{\sim}{A}}}(x) = 1 - \mu_{\underset{\sim}{A}}(x) \tag{1.10}$$

$$\underset{\sim}{A} \subseteq X \implies \mu_{\underset{\sim}{A}}(x) \leq \mu_X(x) \tag{1.11}$$

$$\text{For all } x \in X, \mu_{\emptyset}(x) = 0 \tag{1.12}$$

$$\text{For all } x \in X, \mu_X(x) = 1 \tag{1.13}$$

- The collection of all fuzzy sets and fuzzy subsets on X is denoted as the fuzzy power set $P(\underset{\sim}{X})$.
- De Morgan's principles for classical sets also hold for fuzzy sets, as denoted by the following expressions:

$$\overline{\underset{\sim}{A} \cap \underset{\sim}{B}} = \overline{\underset{\sim}{A}} \cup \overline{\underset{\sim}{B}} \tag{1.14}$$

$$\overline{\underset{\sim}{A} \cup \underset{\sim}{B}} = \overline{\underset{\sim}{A}} \cap \overline{\underset{\sim}{B}} \tag{1.15}$$

- The excluded middle axioms do not hold for fuzzy sets.
- The excluded middle axioms, extended for fuzzy sets, are expressed as

$$\underset{\sim}{A} \cup \overline{\underset{\sim}{A}} \neq X \tag{1.16}$$

$$\underset{\sim}{A} \cap \overline{\underset{\sim}{A}} \neq \emptyset \tag{1.17}$$

1.5.2 Properties of Fuzzy Sets

Fuzzy sets follow the same properties as crisp sets. Because of this fact and because the membership values of a crisp set are a subset of the interval $[0, 1]$, classical sets can be thought of as a special case of fuzzy sets.

2 CLASSICAL RELATIONS AND FUZZY RELATIONS

2.1 CARTESIAN PRODUCT

- An ordered sequence of r elements, written in the form $(a_1, a_2, a_3, \ldots, a_r)$, is called an ***ordered*** $r-$t***uple***; an unordered $r-$ tuple is simply a collection of r elements without restrictions on order.
- For crisp sets $A_1, A_2, \ldots, A_r$, the set of all $r-$tuples $(a_1, a_2, a_3, \ldots, a_r)$, where $a_1 \in A_1,\ a_2 \in A_2,$ and $a_r \in A_r$, is called the ***Cartesian product*** of $A_1, A_2, \ldots, A_r$, and is denoted by $A_1 \times A_2 \times \cdots \times A_r$.
- When all the A_r are identical and equal to A, the Cartesian product $A_1, A_2, \ldots, A_r$ can be denoted as A^r.

Example 3. The elements in two sets A and B are given as $A = \{0, 1\}$ and $B = \{a, b, c\}$. Find $A \times B$, $B \times A$, A^2 .

2.2 CRISP RELATIONS

- A subset of the Cartesian product $A_1 \times A_2 \times \cdots \times A_r$ is called an $r-$***ary relation*** over $A_1, A_2, \ldots, A_r$.
- The most common case is for $r = 2$; in this situation, the relation is a subset of the Cartesian product $A_1 \times A_2$ and this subset of the full Cartesian product is called ***a binary relation*** from A_1 into A_2.
- The Cartesian product of two universes X and Y is determined as

$$X \times Y = \{(x, y) | x \in X,\ y \in Y\}, \tag{2.1}$$

which forms an ordered pair of every $x \in X$ with every $y \in Y$, forming ***unconstrained*** matches between X and Y. That is, every element in universe X is related completely to every element in universe Y.

- The ***strength*** of this relationship between ordered pairs of elements in each universe is measured by the characteristic function, denoted χ, where

$$\chi_{X \times Y}(x, y) = \begin{cases} 1 & (x, y) \in X \times Y \\ 0 & (x, y) \notin X \times Y \end{cases} \tag{2.2}$$

- When the universes, or sets, are finite the relation can be conveniently represented by a matrix, called a ***relation matrix***. An $r-$ary relation can be represented by an $r-$dimensional relation matrix. Hence, binary relations can be represented by two-dimensional matrices.
- For example,

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} a & b & c \end{array} \\ \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \end{array} \tag{2.3}$$

correspond to the ordered pairs of mappings in the relation of the two universes that are defined as $X = \{1, 2, 3\}$ and $Y = \{a, b, c\}$.

- A more general crisp relation, R, exists when matches between elements in two universes are ***constrained***.

Example 4. In many biological models, members of certain species can reproduce only with certain members of another species. Hence, only some elements in two or more universes have a relationship (nonzero) in the Cartesian product. Let the two two-member species, $X = \{1, 2\}$ and $Y = \{a, b\}$. What is the relation matrix for $R = \{(1, a), (2, b)\}$.

- Special cases of the constrained and the unconstrained Cartesian product for sets where $r = 2$ (i.e., for A^2) are called the identity relation and the universal relation, respectively. For example, for $A = \{0, 1, 2\}$ the ***universal relation***, denoted U_A, and the ***identity relation***, denoted I_A, are found to be

$$U_A = \{(0,0), (0,1), (0,2), (1,0), (1,1), (1,2), (2,0), (2,1), (2,2)\}$$
$$I_A = \{(0,0), (1,1), (2,2)\}$$

Example 5. Relations can also be defined for continuous universes. Consider, for example, the continuous relation defined by the following expression:

$$R = \{(x, y) | y \geq 2x,\ x \in X,\ y \in Y\}, \tag{2.4}$$

which is also given in function-theoretic form using the characteristic function as

$$\chi_R(x, y) = \begin{cases} 1 & y \geq 2x \\ 0 & y < 2x \end{cases} \tag{2.5}$$

Show this relaton graphically.

2.2.1 Cardinality of Crisp Relations

- Suppose n elements of the universe X are related (paired) to m elements of the universe Y.
- If the cardinality of X is n_X and the cardinality of Y is n_Y, then the cardinality of the relation, R, between these two universes is $n_{X\times Y} = n_X * n_Y$.
- The cardinality of the power set describing this relation, $P(X \times Y)$, is then

$$n_{P(X\times Y)} = 2^{(n_X n_Y)}. \tag{2.6}$$

2.2.2 Operations on Crisp Relations

Define R and S as two separate relations on the Cartesian universe $X \times Y$, and define the null relation and the complete relation as the relation matrices **O** and **E**, respectively. An example of a 4×4 form of the **O** and **E** matrices is given here:

$$\mathbf{O} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad , \quad \mathbf{E} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \tag{2.7}$$

- The following function-theoretic operations for the two crisp relations (R, S) can now be defined.

$$\text{Union} \quad R \cup S \to \chi_{R\cup S}(x, y) = \max\{\chi_R(x, y), \chi_S(x, y)\} \tag{2.8}$$

$$\text{Intersection} \quad R \cap S \to \chi_{R\cap S}(x, y) = \min\{\chi_R(x, y), \chi_S(x, y)\} \tag{2.9}$$

$$\text{Complement} \quad \overline{R} \to \chi_{\overline{R}}(x, y) = 1 - \chi_R(x, y) \tag{2.10}$$

$$\text{Containment} \quad R \subseteq S \to \chi_R(x, y) \leq \chi_S(x, y) \tag{2.11}$$

$$\text{Identity} \quad \emptyset \to \mathbf{O} \text{ and } X \to \mathbf{E} \tag{2.12}$$

2.2.3 Properties of Crisp Relations

The properties of commutativity, associativity, distributivity, involution, and idempotency all hold for crisp relations just as they do for classical set operations. Moreover, De Morgan's principles and the excluded middle axioms also hold for crisp (classical) relations just as they do for crisp (classical) sets.

2.2.4 Composition

- Let R be a relation that relates, or maps, elements from universe X to universe Y, and let S be a relation that relates, or maps, elements from universe Y to universe Z.
- We can find a relation, T, that relates the same elements in universe X that R contains to the same elements in universe Z that S contains. Thid relation is known as ***composition***.
- There are two common forms of the composition operation: one is called the ***max– min composition*** and the other the ***max–product composition***.
- The max–min composition is defined by the set-theoretic and membership function-theoretic expressions

$$T = R \circ S,$$
$$\chi_T(x, z) = \bigvee_{y\in Y} (\chi_R(x, y) \wedge \chi_S(y, z)) \tag{2.13}$$

- The max–product (sometimes called max–dot) composition is defined by the settheoretic and membership function-theoretic expressions

$$T = R \circ S,$$
$$\chi_T(x,z) = \bigvee_{y \in Y} (\chi_R(x,y) \bullet \chi_S(y,z)) \tag{2.14}$$

Here, the symbol "•" is arithmetic product.

2.3 FUZZY RELATIONS

Fuzzy relation $\underset{\sim}{R}$ is a mapping from the Cartesian space $X \times Y$ to the interval $[0,1]$, where the strength of the mapping is expressed by the membership function of the relation for ordered pairs from the two universes, or $\mu_{\underset{\sim}{R}}(x,y)$.

2.3.1 Cardinality of Fuzzy Relations

Since the cardinality of fuzzy sets on any universe is infinity, the cardinality of a fuzzy relation between two or more universes is also infinity.

2.3.2 Operations on Fuzzy Relations

Let $\underset{\sim}{R}$ and $\underset{\sim}{S}$ be fuzzy relations on the Cartesian space $X \times Y$. Then the following operations apply for the membership values for various set operations:

$$\text{Union} \quad \mu_{\underset{\sim}{R} \cup \underset{\sim}{S}}(x,y) = \max(\mu_{\underset{\sim}{R}}(x,y), \mu_{\underset{\sim}{S}}(x,y) \tag{2.15}$$
$$\text{Intersection} \quad \mu_{\underset{\sim}{R} \cap \underset{\sim}{S}}(x,y) = \min(\mu_{\underset{\sim}{R}}(x,y), \mu_{\underset{\sim}{S}}(x,y)) \tag{2.16}$$
$$\text{Complement} \quad \mu_{\overline{\underset{\sim}{R}}}(x,y) = 1 - \mu_{\underset{\sim}{R}}(x,y) \tag{2.17}$$
$$\text{Containment} \quad \underset{\sim}{R} \subseteq S \implies \mu_{\underset{\sim}{R}}(x,y) \leq \mu_S(x,y) \tag{2.18}$$

2.3.3 Properties of Fuzzy Relations

The properties of commutativity, associativity, distributivity, involution, and idempotency all hold for fuzzy relations. Moreover, De Morgan's principles hold for fuzzy relations just as they do for crisp (classical) relations. The excluded middle axioms for fuzzy relations do not result.

2.3.4 Fuzzy Cartesian Product and Composition

- Because fuzzy relations in general are fuzzy sets, we can define the Cartesian product to be a relation between two or more fuzzy sets.
- Let $\underset{\sim}{A}$ be a fuzzy set on universe X and $\underset{\sim}{B}$ be a fuzzy set on universe Y, then the Cartesian product between fuzzy sets $\underset{\sim}{A}$ and $\underset{\sim}{B}$ willresult in a fuzzy relation $\underset{\sim}{R}$, which is contained within the full Cartesian product space, or

$$\underset{\sim}{A} \times \underset{\sim}{B} = \underset{\sim}{R} \subset X \times Y \tag{2.19}$$

where the fuzzy relation $\underset{\sim}{R}$ has membership function

$$\mu_{\underset{\sim}{R}}(x,y) = \min(\mu_{\underset{\sim}{A}}(x), \mu_{\underset{\sim}{B}}(y)) \tag{2.20}$$

- Suppose $\underset{\sim}{R}$ is a fuzzy relation on the Cartesian space $X \times Y$, $\underset{\sim}{S}$ is a fuzzy relation on $Y \times Z$, and $\underset{\sim}{T}$ is a fuzzy relation on $X \times Z$, then fuzzy max–min composition is defined in terms of the set-theoretic notation and membership function-theoretic notation in the following manner:

$$\underset{\sim}{T} = \underset{\sim}{R} \circ \underset{\sim}{S},$$
$$\mu_{\underset{\sim}{T}}(x,z) = \bigvee_{y \in Y} (\mu_{\underset{\sim}{R}}(x,y) \wedge \mu_{\underset{\sim}{S}}(y,z)) \tag{2.21}$$

- Fuzzy max–product composition is defined in terms of the membership functiontheoretic notation as

$$\mu_{\underset{\sim}{T}}(x,z) = \bigvee_{y \in Y} (\mu_{\underset{\sim}{R}}(x,y) \bullet \mu_{\underset{\sim}{S}}(y,z)) \tag{2.22}$$

- Note that neither crisp nor fuzzy compositions are commutative in general; that is,

$$\underset{\sim}{R} \circ \underset{\sim}{S} \neq \underset{\sim}{S} \circ \underset{\sim}{R} \tag{2.23}$$

2.4 TOLERANCE AND EQUIVALENCE RELATIONS

- Relations can exhibit various useful properties, a few of which are
  - reflexivity,
  - symmetry, and
  - transitivity
- Consider the simple graphs in Figure 2.1. This figure describes a universe of three elements, which are labeled as the vertices of this graph, 1, 2, and 3, or in set notation, $X = \{1, 2, 3\}$. [Note: A relations can be used in graph theory]

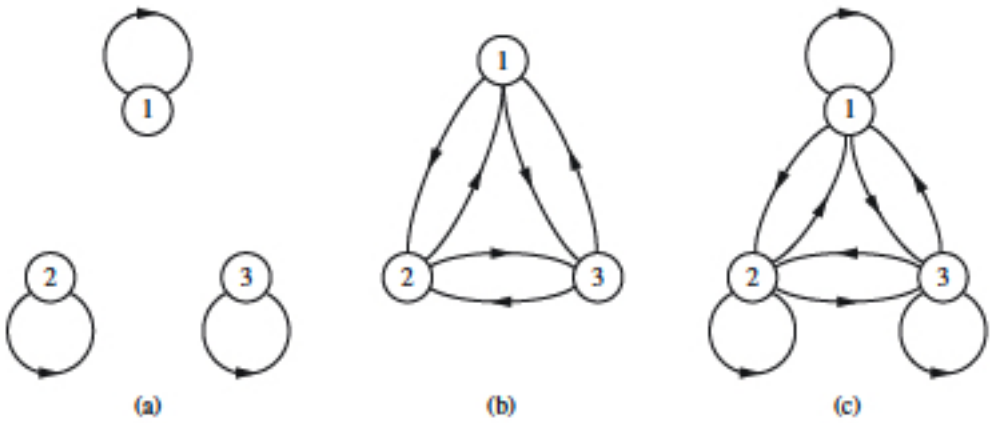


Figure 2.1

- When a relation is reflexive every vertex in the graph originates a single loop, as shown in Figure 2.1(a).
- If a relation is symmetric, then in the graph for every edge pointing (the arrows on the edge lines in Figure 2.1(b) from vertex i to vertex j $(i, j = 1, 2, 3)$, there is an edge pointing in the opposite direction, that is, from vertex j to vertex i.
- When a relation is transitive, then for every pair of edges in the graph, one pointing from vertex i to vertex j and the other from vertex j to vertex k $(i, j, k = 1, 2, 3)$, there is an edge pointing from vertex i directly to vertex k, as seen in Figure 2.1(c) (e.g., an arrow from vertex 1 to vertex 2, an arrow from vertex 2 to vertex 3, and an arrow from vertex 1 to vertex 3).

2.4.1 Crisp Equivalence Relation

- A relation R on a universe X can also be thought of as a relation from X to X.
- The relation R is an equivalence relation if it has the following three properties: (1) reflexivity, (2) symmetry, and (3) transitivity.
- For example, for a matrix relation, the following properties will hold:

$$\text{Reflexivity} \quad (x_i, x_i) \in R \text{ or } \chi_R(x_i, x_i) = 1 \tag{2.24}$$

$$\text{Symmetry} \quad (x_i, x_j) \in R \rightarrow (x_j, x_i) \in R \text{ or } \chi_R(x_i, x_j) = \chi_R(x_j, x_i) \tag{2.25}$$

$$\text{Transitivity} \quad (x_i, x_j) \in R \text{ and } (x_j, x_k) \in R \rightarrow (x_i, x_k) \in R \text{ or } \chi_R(x_i, x_j) \text{ and } \chi_R(x_j, x_k) = 1 \rightarrow \chi_R(x_i, x_k) = 1 \tag{2.26}$$

2.4.2 Crisp Tolerance Relation

- A tolerance relation R (also called a *proximity* relation) on a universe X is a relation that exhibits only the properties of reflexivity and symmetry.
- A tolerance relation, R, can be reformed into an equivalence relation by at most $(n-1)$ compositions with itself, where n is the cardinal number of the set defining R, in this case X, that is,

$$R_1^{n-1} = R_1 \circ R_1 \circ \cdots \circ R_1 = R. \tag{2.27}$$

2.4.3 FUZZY TOLERANCE AND EQUIVALENCE RELATIONS

- A fuzzy relation, $\underset{\sim}{R}$, on a single universe X is also a relation from X to X.
- It is a fuzzy equivalence relation if all three of the following properties for matrix relations define it:

$$\text{Reflexivity} \quad \mu_{\underset{\sim}{R}}(x_i, x_i) = 1 \tag{2.28}$$

$$\text{Symmetry} \quad \mu_{\underset{\sim}{R}}(x_i, x_j) = \mu_{\underset{\sim}{R}}(x_j, x_i) \tag{2.29}$$

$$\text{Transitivity} \quad \mu_{\underset{\sim}{R}}(x_i, x_j) = \lambda_1 \text{ and } \mu_{\underset{\sim}{R}}(x_j, x_k) = \lambda_2 \rightarrow \mu_{\underset{\sim}{R}}(x_i, x_k) = \lambda \text{ where } \lambda \geq \min[\lambda_1, \lambda_2] \tag{2.30}$$

- It can be shown that any fuzzy tolerance relation, $\underset{\sim}{R_1}$, that has properties of reflexivity and symmetry can be reformed into a fuzzy equivalence relation by at most $(n-1)$ compositions, just as a crisp tolerance relation can be reformed into a crisp equivalence relation. That is,

$$\underset{\sim}{R_1}^{n-1} = \underset{\sim}{R_1} \circ \underset{\sim}{R_1} \circ \cdots \circ \underset{\sim}{R_1} = \underset{\sim}{R}. \tag{2.31}$$

2.5 VALUE ASSIGNMENTS

- There are different ways to develop the numerical values that characterize a relation. Few are listed below.
  1. Cartesian product
  2. Closed-form expression
  3. Lookup table
  4. Linguistic rules of knowledge
  5. Similarity methods in data manipulation
- One of the most prevalent forms of determining the values in relations is through manipulations of data, the similarity methods.
- The similarity methods attempt to determine some sort of similar pattern or structure in data through various metrics. There are many of these methods available, but the two most prevalent are discussed here.

2.5.1 Cosine Amplitude

- Consider a collection of data samples, n data samples that forms a data array, $\mathbf{X}$,

$$\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}. \tag{2.32}$$

- Each of the elements, $\mathbf{x}_i$, in the data array $\mathbf{X}$ is itself a vector of length m, that is,

$$\mathbf{x}_i = \{\mathbf{x}_{i_1}, \mathbf{x}_{i_2}, \ldots, \mathbf{x}_{i_m}\}. \tag{2.33}$$

- A pairwise comparison of two data samples, say $\mathbf{x}_i$ and $\mathbf{x}_j$, where the strength of the relationship between data sample $\mathbf{x}_i$ and data sample $\mathbf{x}_j$ is given by the membership value expressing that strength, that is, $r_{ij} = \mu_R(x_i, y_j)$.
- The relation matrix will be of size $n \times n$ and, the matrix will be reflexive and symmetric – hence a tolerance relation.
- The cosine amplitude method is one of the popular similarity methods that calculates r_{ij} in the following manner:

$$r_{ij} = \frac{\left|\sum_{k=1}^{m} x_{ik}x_{jk}\right|}{\sqrt{\left(\sum_{k=1}^{m} x_{ik}^2\right)\left(\sum_{k=1}^{m} x_{jk}^2\right)}} \tag{2.34}$$

where $i, j = 1, 2, \ldots, n$. Note that $0 \leq r_{ij} \leq 1$.

2.5.2 Max–Min Method

- Another popular method, which is computationally simpler than the cosine amplitude method, is known as the *max–min method*.
- It is found through simple min and max operations on pairs of the data points, x_{ij}, and is given as

$$r_{ij} = \frac{\sum_{k=1}^{m} \min(x_{ik}, x_{jk})}{\sum_{k=1}^{m} \max(x_{ik}, x_{jk})} \tag{2.35}$$

where $i, j = 1, 2, \ldots, n$.

2.6 OTHER FORMS OF THE COMPOSITION OPERATION

- Max–min and max–product (also referred to as max–dot) methods of composition of fuzzy relations are the two most commonly used techniques. Many other techniques are mentioned in the literature.
- The following additional methods are among those proposed in the literature for the composition operation $\underset{\sim}{B} = \underset{\sim}{A} \circ \underset{\sim}{R}$, where $\underset{\sim}{A}$ is the input, or antecedent defined on the universe X, $\underset{\sim}{B}$ is the output, or consequent defined on universe $\underset{\sim}{Y}$, and $\underset{\sim}{R}$ is a fuzzy relation characterizing the relationship between specific inputs (x) and specific outputs (y):

$$\text{min-max} \quad \mu_{\underset{\sim}{B}}(y) = \min_{x \in X}\{\max[\mu_{\underset{\sim}{A}}(x), \mu_{\underset{\sim}{R}}(x, y)]\} \tag{2.36}$$

$$\text{max-max} \quad \mu_{\underset{\sim}{B}}(y) = \max_{x \in X}\{\max[\mu_{\underset{\sim}{A}}(x), \mu_{\underset{\sim}{R}}(x, y)]\} \tag{2.37}$$

$$\text{min-min} \quad \mu_{\underset{\sim}{B}}(y) = \min_{x \in X}\{\min[\mu_{\underset{\sim}{A}}(x), \mu_{\underset{\sim}{R}}(x, y)]\} \tag{2.38}$$

$$\text{max-average} \quad \mu_{\underset{\sim}{B}}(y) = \frac{1}{2} \max_{x \in X}[\mu_{\underset{\sim}{A}}(x) + \mu_{\underset{\sim}{R}}(x, y)] \tag{2.39}$$

$$\text{sum-product} \quad \mu_{\underset{\sim}{B}}(y) = f\Big(\sum_{x \in X}[\mu_{\underset{\sim}{A}}(x).\mu_{\underset{\sim}{R}}(x, y)]\Big) \tag{2.40}$$

where $f(.)$ is a logistic function (such as a sigmoid or a step function) that limits the value of the function within the interval $[0, 1]$.

- This composition method is commonly used in applications of artificial neural networks for mapping between parallel layers in a multilayer network.

3 PROPERTIES OF MEMBERSHIP FUNCTIONS, FUZZIFICATION, AND DEFUZZIFICATION

3.1 FEATURES OF THE MEMBERSHIP FUNCTION

NOTE: For purposes of simplicity, the functions shown in the figures will all be continuous, but the terms apply equally for both discrete and continuous fuzzy sets.

- The ***core*** of a membership function for some fuzzy set $\underset{\sim}{A}$ is defined as that region of the universe that is characterized by complete and full membership in the set $\underset{\sim}{A}$. That is, the core comprises those elements x of the universe such that $\mu_{\underset{\sim}{A}}(x) = 1$.
- The ***support*** of a membership function for some fuzzy set $\underset{\sim}{A}$ is defined as that region of the universe that is characterized by nonzero membership in the set $\underset{\sim}{A}$. That is, the support comprises those elements x of the universe such that $\mu_{\underset{\sim}{A}}(x) > 0$.
- The boundaries of a membership function for some fuzzy set $\underset{\sim}{A}$ are defined as that region of the universe containing elements that have a nonzero membership but not complete membership. That is, the boundaries comprise those elements x of the universe such that $0 < \mu_{\underset{\sim}{A}}(x) < 1$. These elements of the universe are those with some degree of fuzziness, or only partial membership in the fuzzy set $\underset{\sim}{A}$.

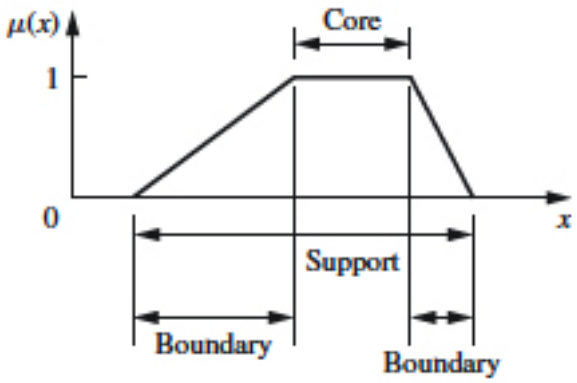


Figure 3.1

- A *normal* fuzzy set is one whose membership function has at least one element x in the universe whose membership value is unity. In fuzzy sets, where one and only one element has a membership equal to one, the element is typically referred to as the *prototype* of the set, or the *prototypical element*.

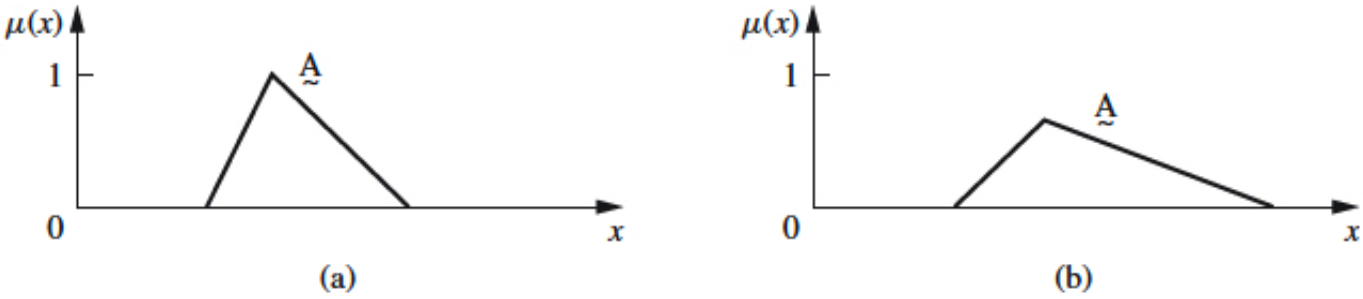


Figure 3.2: Fuzzy sets that are (a) normal and (b) subnormal

- A *convex* fuzzy set is described by a membership function whose membership values are strictly monotonically increasing [1], or whose membership values are strictly monotonically decreasing[2], or whose membership values

[1] A function is called monotonically increasing, if for all x and y such that $x \leq y$ one has $f(x) \leq f(y)$.
[2] A function is called monotonically decreasing, if for all x and y such that $x \leq y$ one has $f(x) \geq f(y)$.

are strictly monotonically increasing then strictly monotonically decreasing with increasing values for elements in the universe. That is, if, for any elements x, y, and z in a fuzzy set $\underset{\sim}{A}$, the relation x < y <z implies that $\mu_{\underset{\sim}{A}}(y) \geq \min[\mu_{\underset{\sim}{A}}(x), \mu_{\underset{\sim}{A}}(z)]$, then $\underset{\sim}{A}$ is said to be a convex fuzzy set.

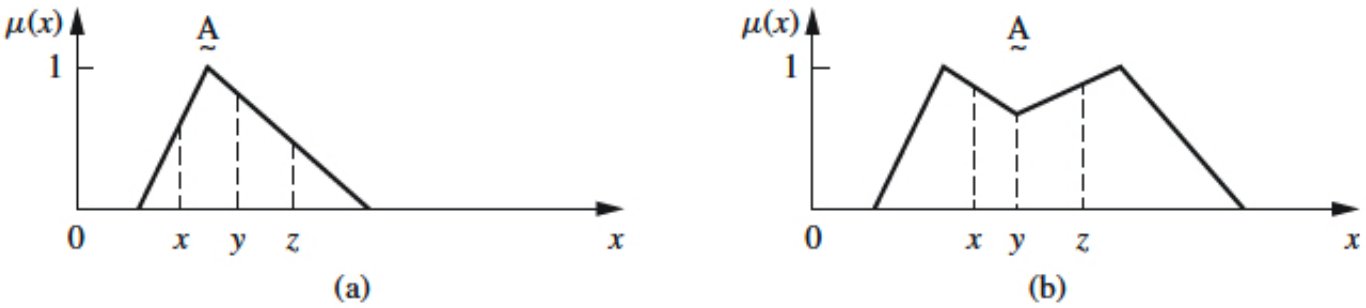


Figure 3.3: (a) Convex, normal fuzzy set and (b) nonconvex, normal fuzzy set

- A special property of two convex fuzzy sets, say $\underset{\sim}{A}$ and $\underset{\sim}{B}$, is that the intersection of these two convex fuzzy sets is also a convex fuzzy set, as shown in Figure 3.4. That is, for $\underset{\sim}{A}$ and $\underset{\sim}{B}$, which are both convex, $\underset{\sim}{A} \cap \underset{\sim}{B}$ is also convex.

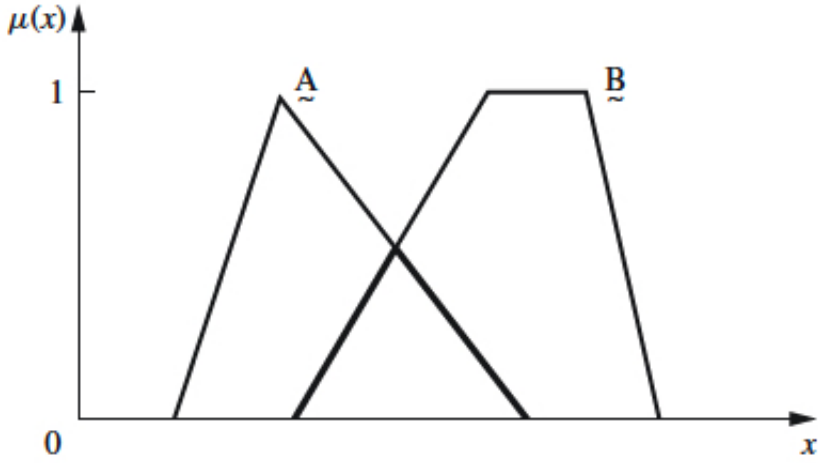


Figure 3.4

- The *crossover points* of a membership function are defined as the elements in the universe for which a particular fuzzy set $\underset{\sim}{A}$ has values equal to 0.5, that is, for which $\mu_{\underset{\sim}{A}}(x) = 0.5$.
- The *height* of a fuzzy set $\underset{\sim}{A}$ is the maximum value of the membership function, that is, $\text{hgt}(\underset{\sim}{A}) = \max\{\mu_{\underset{\sim}{A}}(x)\}$. If the $\text{hgt}(\underset{\sim}{A}) < 1$, the fuzzy set is said to be subnormal.
- If $\underset{\sim}{A}$ is a convex single-point normal fuzzy set defined on the real line, then $\underset{\sim}{A}$ is often termed a fuzzy number.

3.2 VARIOUS FORMS

- The most common forms of membership functions are those that are normal and convex. However, many operations on fuzzy sets, hence operations on membership functions, result in fuzzy sets that are subnormal and non convex.
- Membership functions can be symmetrical or asymmetrical. They are typically defined on one-dimensional universes, but they certainly can be described on multidimensional (or $n-$dimensional) universes.
- Fuzzy sets of the types depicted in Figure 3.2 are by far the most common ones encountered in practice; they are described as *ordinary membership functions*.

- However, several other types of fuzzy membership functions have been proposed as *generalized membership functions*. The primary reason for considering other types of membership functions is that the values used in developing ordinary membership functions are often overly precise. They require that each element x of the universe on which the fuzzy set $\underset{\sim}{A}$ is defined be assigned a specific membership value, $\mu_{\underset{\sim}{A}}(x)$.
  - Suppose the level of information is not adequate to specify membership functions with the given precision. For example, we may know only the upper and lower bounds of membership grades for each element of the universe for a fuzzy set. Such a fuzzy set would be described by an *interval-valued membership function*.

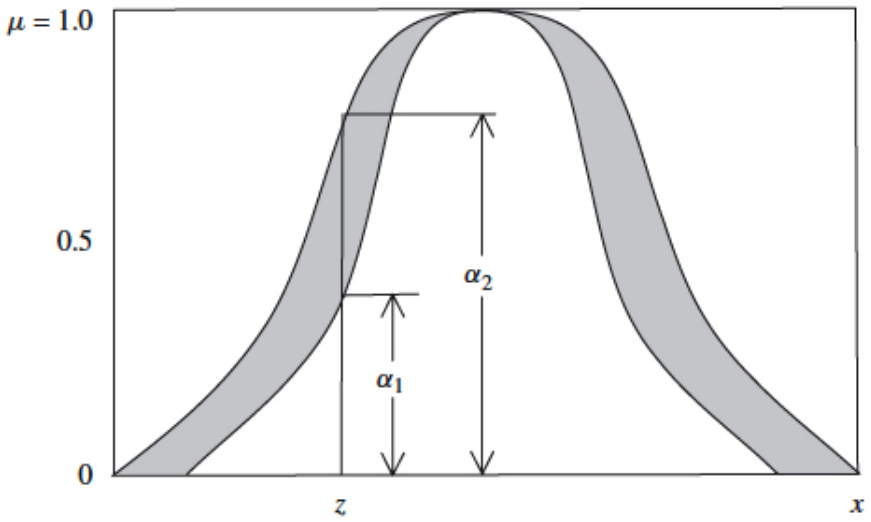


Figure 3.5

In this figure, for a particular element, $x = z$, the membership in a fuzzy set $\underset{\sim}{A}$, that is, $\mu_{\underset{\sim}{A}}(z)$, would be expressed by the membership interval $[\alpha_1, \alpha_2]$. Interval-valued fuzzy sets can be generalized further by allowing their intervals to become fuzzy. Each membership interval then becomes an ordinary fuzzy set. This type of membership function is referred to in the literature as a *type-2 fuzzy set* .

3.3 FUZZIFICATION

- Fuzzification is the process of making a crisp quantity fuzzy. We do this by simply recognizing that many of the quantities that we consider to be crisp and deterministic are actually not deterministic at all; they carry considerable uncertainty. If the form of uncertainty happens to arise because of imprecision, ambiguity, or vagueness, then the variable is probably fuzzy and can be represented by a membership function.

3.4 DEFUZZIFICATION

- Defuzzification is the process of conversion of a fuzzy quantity into a precise quantity.

3.4.1 DEFUZZIFICATION TO CRISP SETS

$\lambda-$ cuts for fuzzy sets (Alpha-Cuts)

- Considering a fuzzy set $\underset{\sim}{A}$, then define a lambda-cut set, A_λ, where $0 \leq \lambda \leq 1$. The set A_λ is a crisp set called the lambda (λ)-cut (or alpha-cut) set of the fuzzy set $\underset{\sim}{A}$, where $A_\lambda = \{x | \mu_{\underset{\sim}{A}}(x) \geq \lambda\}$.
- Any particular fuzzy set $\underset{\sim}{A}$ can be transformed into an infinite number of $\lambda-$cut sets, because there are an infinite number of values λ on the interval $[0, 1]$.

The properties of $\lambda-$ cut sets

1. $(\underset{\sim}{A} \cup \underset{\sim}{B})_\lambda = A_\lambda \cup B_\lambda$
2. $(\underset{\sim}{A} \cap \underset{\sim}{B})_\lambda = A_\lambda \cap B_\lambda$

3. $(\tilde{A})_\lambda \neq \overline{A}_\lambda$ except when $\lambda = 0.5$
4. For any $\lambda \leq \alpha$, where $0 \leq \alpha \leq 1$, it is true that $A_\alpha \subseteq A_\lambda$, where $A_0 = X$

NOTE: The following definitions can be provided for a convex fuzzy set $\tilde{A}$.

- The core of $\tilde{A}$ is the $\lambda = 1$ cut set, A_1.
- The support of $\tilde{A}$ is the $\lambda-$cut set A_{0_+}, where $\lambda = 0+$, or symbolically, $A_{0_+} = \{x|\mu_{\tilde{A}}(x) > 0\}$. The intervals $[A_{0_+}, A_1]$ form the boundaries of the fuzzy set $\tilde{A}$, that is, those regions that have membership values between 0 and 1 (exclusive of 0 and 1): that is, for $0 < \lambda < 1$.

$\lambda-$ cuts for fuzzy relation

- Consider a fuzzy relation $\tilde{R}$, where each row of the relational matrix is considered a fuzzy set, that is, the j th row in $\tilde{R}$ represents a discrete membership function for a fuzzy set,$\overset{\sim}{R_j}$.
- A fuzzy relation can be converted to a crisp relation in the following manner. Let us define

$$R_\lambda = \{(x, y)|\mu_{\tilde{R}}(x, y) \geq \lambda\} \tag{3.1}$$

as a $\lambda-$cut relation of the fuzzy relation $\tilde{R}$.

3.4.2 Methods of Defuzzification To Scalars

- Defuzzification is the conversion of a fuzzy quantity to a precise quantity, just as fuzzification is the conversionof a precise quantity to a fuzzy quantity.
- The output of a fuzzy process can be the logical union of two or more fuzzy membership functions defined on the universe of discourse of the output variable.
- For example, suppose a fuzzy output comprises two parts: (1) $\overset{\sim}{C_1}$, a trapezoidal shape (Figure 3.6(a)) and (2) $\overset{\sim}{C_2}$, a triangular membership shape (Figure 3.6(b)).The union of these two membership functions, that is, $\tilde{C} = \overset{\sim}{C_1} \cup \overset{\sim}{C_2}$, involves the max operator, which graphically is the outer envelope of the two shapes shown in Figure 3.6(a) and (b); the resulting shape is shown in Figure 3.6(c).

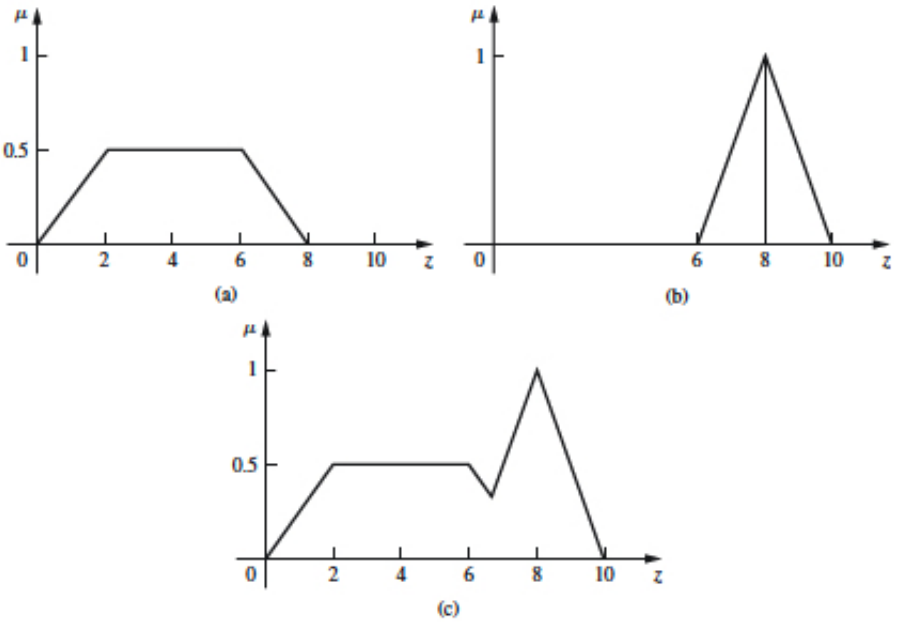


Figure 3.6

- A general fuzzy output process can involve many output parts (more than two), and the membership function-representing each part of the output can have shapes other than triangles and trapezoids. In general, we can have

$$\underset{\sim}{C_k} = \cup_{i=1}^{k} \underset{\sim}{C_i} = \underset{\sim}{C} \tag{3.2}$$

- Among the many methods that have been proposed in the literature in recent years,few are described here for defuzzifying fuzzy output functions (membership functions).

1. *Max membership principl*e: Also known as the *height method*, this scheme is limited to peaked output functions. This method is given by the algebraic expression

$$\mu_{\underset{\sim}{C}}(z^*) \geq \mu_{\underset{\sim}{C}}(z), \tag{3.3}$$

for all $z \in Z$, where z^* is the defuzzified value.

2. Centroid method

$$z^* = \frac{\int \mu_{\underset{\sim}{C}}(z) \cdot z \; dz}{\int \mu_{\underset{\sim}{C}}(z) \; dz} \tag{3.4}$$

3. Weighted average method: The weighted average method is the most frequently usedin fuzzy applications since it is one of the more computationally efficient methods.Unfortunately, it is usually restricted to symmetrical output membership functions.

$$z^* = \frac{\sum \mu_{\underset{\sim}{C}}(\bar{z}) \cdot \bar{z}}{\sum \mu_{\underset{\sim}{C}}(\bar{z})} \tag{3.5}$$

where $\sum$ denotes the algebraic sum and where $\bar{z}$ is the centroid of each symmetric membership function. The weighted average method is formed by weighting each membership function in the output by its respective maximum membership value. As an example, the two functions shown in Figure 3.7 would result in the following general form for the defuzzified value:

$$z^* = \frac{a(0.5) + b(0.9)}{0.5 + 0.9} \tag{3.6}$$

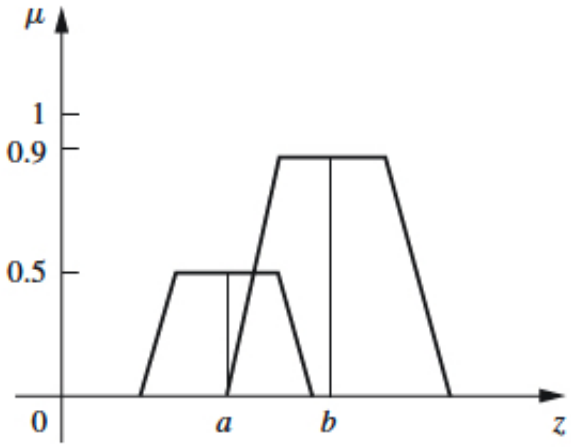


Figure 3.7

4. Mean max membership: This method (also called middle-of-maxima) is closely related to the first method, except that the locations of the maximum membership can be non unique (i.e., the maximum membership can be a plateau rather than a single point).

$$z^* = \frac{a+b}{2} \tag{3.7}$$

where a and b are as defined in following Figure 3.8.

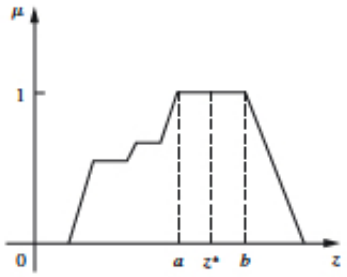


Figure 3.8

5. Center of sums: This is faster than many defuzzification methods that are currently inuse, and the method is not restricted to symmetric membership functions. This processinvolves the algebraic sum of individual output fuzzy sets, say $\underset{\sim}{C_1}$ and $\underset{\sim}{C_2}$, instead oftheir union. Two drawbacks to this method are that the intersecting areas are addedtwice, and the method also involves finding the centroids of the individual membershipfunctions. The defuzzified value z^* is given as follows:

$$z^* = \frac{\sum_{k=1}^{n} \mu_{\underset{\sim}{C_k}}(z) \int_z \bar{z}\ dz}{\sum_{k=1}^{n} \mu_{\underset{\sim}{C_k}}(z) \int_z dz} \tag{3.8}$$

where the symbol $\bar{z}$ is the distance to the centroid of each of the respective membership functions.

This method is similar to the weighted average method, except thatin the center of sums method the weights are the areas of the respective membershipfunctions whereas in the weighted average method the weights are individualmembership values.

6. Center of largest area: If the output fuzzy set has at least two convex subregions, thenthe center of gravity (i.e., z^* is calculated using the centroid method, Equation 3.4) of the convex fuzzy subregion with the largest area is used to obtain the defuzzified value z^* of the output.

$$z^* = \frac{\int \mu_{\underset{\sim}{C_m}}(z) z\ dz}{\int \mu_{\underset{\sim}{C_m}}(z)\ dz} \tag{3.9}$$

where $\underset{\sim}{C_m}$ is the convex subregion that has the largest area making up $\underset{\sim}{C_k}$. This condition applies in the case when the overall output $\underset{\sim}{C_k}$ is non-convex. And, in the case when $\underset{\sim}{C_k}$ is convex, z^* is the same quantity as determined by the centroid method or the center of largest area method (because then there is only one convex region).

4 Logic

4.1 Classical logic: propositions and truth

Definition 4.1: Simple Proposition

In classical logic, a *simple proposition* P is a linguistic (declarative) statement contained within a universe of elements, say X, that can be identified as a collection of elements in X which are *strictly true* or *strictly false*. Hence a proposition P is a *set*: the truth values for all elements in the set are either all true or all false. The veracity (truth or falsity) of an element in P is assigned a binary truth value $T(P) \in \{0, 1\}$ (false $= 0$, true $= 1$).

Let U denote the **universe of discourse** (the set of all elements or propositions under consideration). Truth value assignment can be formalized as a valuation mapping:

$$T \colon U \longrightarrow \{0, 1\} \tag{4.1}$$

where $T(u) = 1$ denotes a true state and $T(u) = 0$ denotes a false state.

For a logical predicate or proposition $P(u)$ evaluated over $u \in U$:

- **Universal Set (U):** The foundational domain containing all instances u.
- **Proposition (P):** A statement function $P(u)$ mapping elements $u \in U$ to binary values $\{\text{True}, \text{False}\}$.
- **Truth Set ($T(P)$):** The subset of elements in U for which $P(u)$ evaluates to true:

$$T(P) = \{u \in U \mid P(u) \text{ is True}\}$$

- **Falsity Set ($F(P)$):** The complement of the truth set within U, containing elements for which $P(u)$ evaluates to false:

$$F(P) = \{u \in U \mid P(u) \text{ is False}\} = \overline{T(P)} = U \setminus T(P)$$

Key Takeaway: The truth set $T(P)$ and falsity set $F(P)$ form a complete partition of the universe U:

$$T(P) \cup F(P) = U \quad \text{and} \quad T(P) \cap F(P) = \emptyset$$

As with sets, we postulate the boundary conditions of truth values: for a universe Y and the null set $\emptyset$,

$$T(Y) = 1 \qquad \text{and} \qquad T(\emptyset) = 0. \tag{4.2}$$

Mutual exclusivity

Two propositions P and Q are said to be **mutually exclusive** if their truth sets are disjoint:

$$T(P) \cap T(Q) = \varnothing$$

In this case, the truth of one proposition strictly implies the falsity of the other:

$$P \implies \overline{Q} \quad \text{and} \quad Q \implies \overline{P}$$

4.2 Logical connectives and compound propositions

Two simple propositions P and Q on the same universe of discourse can be combined with five *logical connectives* to form new (compound) propositions:

Connective	Symbol	Reading
disjunction	$\vee$	logical *or* (inclusive or)
conjunction	$\wedge$	logical *and*
negation	$-$	logical *not*
implication	$\implies$	*implies* (if… then)
equivalence	$\iff$	*if and only if* (dual implication)

Remark (Inclusive vs. Exclusive OR)**.** *In formal logic, the operator $\vee$ denotes the **inclusive OR**: the compound proposition $P \vee Q$ is true if at least one of the component propositions is true, including the case where both are true.*

*In contrast, natural language frequently uses an **exclusive OR** (XOR). For instance, a menu choice of "soup or salad" allows one option or the other, but explicitly excludes selecting both:*

$$T(P \vee Q) = T(P) \cup T(Q) \quad \textit{(Inclusive OR)}$$

Logical Equivalence: The logical equivalence connective arises from **dual implication**. For any two propositions P and Q:

$$\text{if } P \implies Q \text{ and } Q \implies P, \text{ then } P \iff Q$$

Propositions as Set Membership: Let A and B be subsets defined over a universe X (isomorphic to the universe of propositions U). We establish a **propositional calculus** by defining indicator propositions for set membership for an element $x \in X$:

$$P : \text{the proposition that } x \in A$$
$$Q : \text{the proposition that } x \in B$$

Compound Propositions: A **compound proposition** is formed by applying logical connectives to simple propositions. For membership propositions P and Q, compound propositions map to set operations on A and B through their truth values:

The Five Connectives in Truth-Value Form

$$\textit{Disjunction:} \quad P \vee Q: \ x \in A \text{ or } x \in B, \qquad T(P \vee Q) = \max\big(T(P), T(Q)\big), \tag{4.3}$$

$$\textit{Conjunction:} \quad P \wedge Q: \ x \in A \text{ and } x \in B, \qquad T(P \wedge Q) = \min\big(T(P), T(Q)\big), \tag{4.4}$$

$$\textit{Negation:} \quad \overline{P}: \ x \notin A, \qquad T(\overline{P}) = 1 - T(P), \tag{4.5}$$

$$\textit{Implication:} \quad P \implies Q: \ x \notin A \text{ or } x \in B, \qquad T(P \implies Q) = T(\overline{A} \cup B), \tag{4.6}$$

$$\textit{Equivalence:} \quad P \iff Q, \qquad T(P \iff Q) = \begin{cases} 1, & T(P) = T(Q), \\ 0, & T(P) \neq T(Q). \end{cases} \tag{4.7}$$

4.3 Logical Implication and Set Operations

In propositional logic, the statement **"If P, then Q"** (denoted symbolically as $P \implies Q$ or $P \to Q$) is known as a *conditional statement* or *implication*. It states that whenever proposition P is true, proposition Q must also be true.

Terminology:

- P: **Antecedent** (or Hypothesis) — the condition being assumed.
- Q: **Consequent** (or Conclusion) — the resulting outcome.

The implication $P \implies Q$ is logically equivalent to the set union of the complement of A and the set B:

$$(P \implies Q) \ \equiv \ (\overline{A} \cup B) \ \equiv \ (\text{"not in } A\text{" or "in } B\text{"}), \tag{4.8}$$

where $\overline{A}$ (or $X \setminus A$) denotes the set complement. The corresponding valuation maps directly to the max operator for fuzzy/binary logic evaluations:

$$T(P \implies Q) = T(\overline{P} \vee Q) = \max\Big(T(\overline{P}), T(Q)\Big). \tag{4.9}$$

Linguistically, $P \implies Q$ holds true whenever $x \notin A$ or $x \in B$.

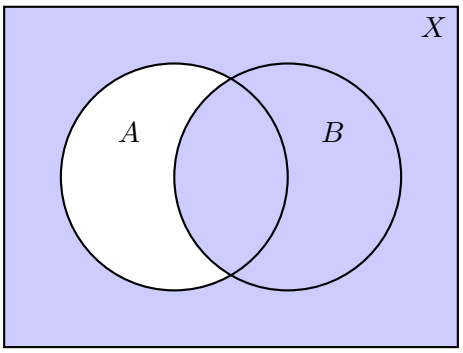


Figure 1: Venn diagram for $\overline{A} \cup B$ (shaded region).

The implication $P \implies Q$ fails if and only if $x \in A$ and $x \notin B$—that is, precisely on the set difference $A \setminus B$.

The truth value of $P \implies Q$ as a function of the truth values of p and q is defined by the following truth table:

Table 1: Truth table for implication

P	Q	$P \implies Q$	**Interpretation**
T	**T**	**T**	p is true and q occurs. The implication holds.
T	**F**	**F**	p is true but q fails to occur. The implication is **false**.
F	**T**	**T**	p is false; q occurring anyway does not violate the promise.
F	**F**	**T**	p is false; q not occurring does not violate the promise.

Vacuous Truth

Notice that $P \implies Q$ is **false if and only if** P is true and Q is false. When P is false, $P \implies Q$ evaluates to **True** regardless of the value of Q. This is known as *vacuous truth*.

4.4 Truth table for two propositions

With two propositions P and Q, each taking one of two truth values, there are $2^2 = 4$ propositional situations (Table 2); the last five columns are computed with (4.3)–(4.7) and (4.9).

Table 2: Truth table for various compound propositions

P	Q	$\overline{P}$	$P \vee Q$	$P \wedge Q$	$P \implies Q$	$P \iff Q$
T (1)	T (1)	F (0)	T (1)	T (1)	T (1)	T (1)
T (1)	F (0)	F (0)	T (1)	F (0)	F (0)	F (0)
F (0)	T (1)	T (1)	T (1)	F (0)	T (1)	F (0)
F (0)	F (0)	T (1)	F (0)	F (0)	T (1)	T (1)

4.5 Implication between two different universes

Suppose the implication involves two *different* universes: P is described by set A on universe X, and Q by set B on universe Y. Then $P \implies Q$ is represented in set-theoretic terms by the *relation* R on the Cartesian product $X \times Y$, where R is defined as

$$R = (A \times B) \cup (\overline{A} \times Y) \;\equiv\; \text{IF } A, \text{ THEN } B, \tag{4.10}$$

That is,

$$P \implies Q: \quad \text{IF } x \in A \; (x \in X,\; A \subset X), \text{ THEN } y \in B \; (y \in Y,\; B \subset Y),$$

equivalently $P \implies Q \equiv \overline{A} \cup B$.

Figure 2 shows the truth domain of this rule in the Cartesian space $X \times Y$.

Another compound proposition in linguistic rule form is

$$\text{IF } A, \text{ THEN } B, \text{ ELSE } C, \tag{4.11}$$

and this compound proposition could be expressed as

$$\text{IF } A, \text{ THEN } B \text{ and IF } \overline{A}, \text{ THEN } C.$$

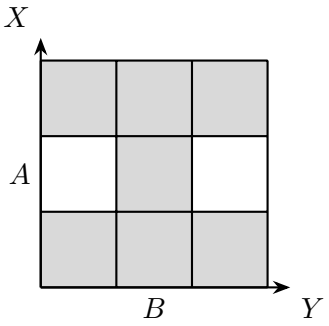


Figure 2: The Cartesian space showing the implication IF A, THEN B.

In classical logic this rule has the form

$$(P \implies Q) \wedge (P \implies S), \tag{4.12}$$

with

$$P : x \in A\,,\; A \subset X$$
$$Q : y \in B\,,\; B \subset Y$$
$$S : y \in C\,,\; C \subset Y$$

Its set-theoretic equivalent is the relation on $X \times Y$,

$$\text{IF } A\text{, THEN } B\text{, ELSE } C \;\equiv\; (A \times B) \cup (\overline{A} \times C) \;=\; R. \tag{4.13}$$

Figure 3 shows the truth domain for the particular case $B \cap C = \varnothing$.

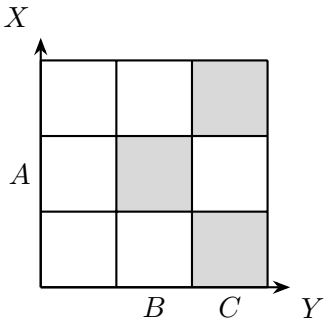


Figure 3: Truth domain for IF A, THEN B, ELSE C.

4.6 Tautologies

Definition 4.2: Tautology

A **tautology** is a compound proposition that evaluates to **true** under all possible truth assignments of its constituent propositions. Because tautologies are logically valid by structure alone, they form the foundation for deductive reasoning, mathematical proofs, and inference engines in expert systems.

Some common tautologies are as follows:

$$(\overline{A} \cup A) \iff X,$$
$$AUX; \overline{A}UX \iff X$$
$$\big(A \wedge (A \implies B)\big) \implies B \qquad (\textit{modus ponens}), \tag{4.14}$$
$$\big(\overline{B} \wedge (A \implies B)\big) \implies \overline{A} \qquad (\textit{modus tollens}). \tag{4.15}$$

Table 3: Truth table for modus tollens

A	B	$\overline{A}$	$\overline{B}$	$A \implies B$	$\big(\overline{B} \wedge (A \implies B)\big)$	$\big(\overline{B} \wedge (A \implies B)\big) \implies \overline{A}$
0	0	1	1	1	1	1
0	1	1	0	1	0	1
1	0	0	1	0	0	1
1	1	0	0	1	0	1

4.7 Contradictions

Definition 4.3: Contradiction

A **contradiction** (or absurdity) is a compound proposition that evaluates to **false** under all possible truth assignments of its constituent simple propositions.

Simple contradictions are

$$(B \cap \overline{B}), \qquad (A \cap \varnothing).$$

4.8 Equivalence and Related Forms

Two propositions P and Q are **equivalent**, denoted $P \iff Q$, if and only if they share the same truth value—that is, when both P and Q are true or both are false.

The equivalence $P \iff Q$ is a **tautology** precisely when P and Q have identical truth sets:

$$T(P) = T(Q)$$

In set-theoretic terms, this means the truth regions $T(P)$ and $T(Q)$ in a Venn diagram coincide completely.

Example 4.1. *Consider the propositions defined over the universe of integers $x \in \mathbb{Z}$:*

$$P\colon \text{“}x \textit{ is an even integer}\text{”}$$
$$Q\colon \text{“}x^2 \textit{ is an even integer}\text{”}$$

The propositions P and Q are logically equivalent ($P \iff Q$) because an integer x is even if and only if its square x^2 is even. For any given integer x, P and Q are either both true or both false, meaning:

$$T(P) = T(Q)$$

Example 4.2. *In-Class Discussion*

4.9 Contrapositive, Converse, Inverse, and Duals

Suppose a conditional proposition R has the form $P \implies Q$. Then we define:

- **Contrapositive:** $\overline{Q} \implies \overline{P}$
- **Converse:** $Q \implies P$
- **Inverse:** $\overline{P} \implies \overline{Q}$

Theorem 4.1 (Equivalences among Related Forms)**.** *For any conditional proposition R:*

1. *A proposition and its contrapositive are logically equivalent:*

$$(P \implies Q) \iff (\overline{Q} \implies \overline{P})$$

2. *The converse and the inverse of R are logically equivalent:*

$$(Q \implies P) \iff (\overline{P} \implies \overline{Q})$$

Definition 4.4: Duality Principle

The **dual** of a compound proposition (containing only connectives $\wedge, \vee$ and constants $0, 1$) is obtained by interchanging:

$$\wedge \longleftrightarrow \vee, \qquad 0 \longleftrightarrow 1, \qquad (\text{or } A \longleftrightarrow \overline{A})$$

Principle of Duality: If a logical identity S is universally valid, then its dual S^* is also universally valid.

4.10 Exclusive OR and Exclusive NOR

The **exclusive OR** (XOR, denoted $\oplus$) arises frequently in natural language and human reasoning. For instance, traveling "by plane or boat" implies taking one mode of transit or the other, but not both simultaneously. Unlike the inclusive logical OR ($\vee$), the exclusive OR excludes the intersection of the two sets.

For propositions P and Q, the exclusive OR is defined algebraically as:

$$P \oplus Q = (P \wedge \overline{Q}) \vee (\overline{P} \wedge Q) \tag{4.16}$$

Set-theoretically, this corresponds to the symmetric difference $A \Delta B = (A \setminus B) \cup (B \setminus A)$, represented graphically by the two disjoint crescents of a Venn diagram.

Table 4: Truth table for Exclusive OR (XOR).

P	Q	$P \oplus Q$
1	1	0
1	0	1
0	1	1
0	0	0

Table 5: Truth table for Exclusive NOR (XNOR).

P	Q	$P \odot Q$
1	1	1
1	0	0
0	1	0
0	0	1

The **exclusive NOR** (XNOR, denoted $\odot$) is the negation of the exclusive OR:

$$P \odot Q = \overline{(P \oplus Q)} = (P \wedge Q) \vee (\overline{P} \wedge \overline{Q})$$

As shown in Table 5, XNOR evaluates to true if and only if P and Q have identical truth values. Thus, it represents logical equivalence:

$$P \odot Q \iff (P \iff Q)$$

4.11 Logical Proofs

Logic involves the use of inference in everyday reasoning as well as in mathematics, where theorems are proven to build foundations for formal solution procedures. Given a set of hypotheses, the process of **inference** allows us to deduce new facts from established premises.

In formal logic, we seek to determine whether a compound implication of the form

$$(P_1 \wedge P_2 \wedge \cdots \wedge P_n) \implies Q \tag{4.17}$$

is universally true—that is, whether the implication is a **tautology**.

Steps for Verifying a Deductive Argument

To prove the validity of a compound linguistic argument, follow these four steps:

1. **Formulate:** State the full compound proposition in natural language.
2. **Decompose:** Break the argument down into its constituent atomic propositions $(P_1, P_2, \ldots, Q)$.
3. **Algebraize:** Translate the natural language statement into formal logic using connectives $(\wedge, \vee, -, \implies)$.
4. **Evaluate:** Construct a truth table or use Boolean algebra to establish whether the resulting expression is a tautology.

Example 4.3.

When a direct proof of an implication is difficult to construct, an **indirect proof** (or *proof by contradiction*) provides a powerful alternative.

This technique exploits the logical equivalence between an implication and the impossibility of its counterexample:

$$(P \implies Q) \quad \iff \quad \overline{(P \wedge \overline{Q})}$$

Thus, to prove that a compound statement with multiple hypotheses

$$(P_1 \wedge P_2 \wedge \cdots \wedge P_n) \implies Q \tag{4.18}$$

is a **tautology**, it is sufficient to show that combining all hypotheses with the negation of the conclusion:

$$P_1 \wedge P_2 \wedge \cdots \wedge P_n \wedge \overline{Q} \tag{4.19}$$

yields a **contradiction** (an expression that evaluates to false under all truth assignments).

Example 4.4.

4.12 Deductive Inferences with Crisp Relations

The *modus ponens* deduction serves as the primary mechanism for making inferences in rule-based systems. A typical conditional rule determines whether an antecedent (cause or action) implies a consequent (effect or reaction).

Consider a rule of the form **IF** A**, THEN** B, where $A \subseteq X$ and $B \subseteq Y$. This rule translates into a binary relation R defined on the Cartesian product $X \times Y$:

$$R = (A \times B) \cup (\overline{A} \times Y)$$

Given a new antecedent $A' \subseteq X$, modus ponens infers a corresponding consequent $B' \subseteq Y$ through relational composition:

$$B' = A' \circ R = A' \circ \big((A \times B) \cup (A^c \times Y)\big) \tag{4.20}$$

where $\circ$ denotes the max-min (or boolean relational) composition.

For a compound conditional rule of the form **IF** A**, THEN** B**, ELSE** C, the relation is defined as:

$$R = (A \times B) \cup (\overline{A} \times C)$$

Depending on how a new antecedent A' relates to A, three distinct cases emerge:

$$\begin{array}{ll} \text{IF } A' \subseteq A, & \text{THEN } y \in B, \\ \text{IF } A' \subseteq \overline{A}, & \text{THEN } y \in C, \\ \text{IF } A' \cap A \neq \emptyset \text{ and } A' \cap \overline{A} \neq \emptyset, & \text{THEN } y \in B \cup C \end{array} \tag{4.21}$$

In characteristic function notation $\chi(\cdot) \in \{0, 1\}$, the **IF** A**, THEN** B relation is given by:

$$\chi_R(x, y) = \max\Big(\min\big(\chi_A(x), \chi_B(y)\big), \min\big(1 - \chi_A(x), 1\big)\Big) \tag{4.22}$$

Example 4.5.

The compound rule **IF** A**, THEN** B**, ELSE** C is evaluated using the characteristic function:

$$\chi_R(x, y) = \max\Big(\min\big(\chi_A(x), \chi_B(y)\big), \min\big(1 - \chi_A(x), \chi_C(y)\big)\Big) \tag{4.23}$$

Example 4.6.

5 Fuzzy logic

5.1 Introduction

In real-world physical systems, human cognitive processes rarely exhibit clear-cut, binary boundaries. Engineering systems deal routinely with imprecise measurements, linguistic terms (such as *"excessive pressure"*, *"high temperature"*, or *"rapid acceleration"*), and incomplete information. Applying crisp thresholds to continuously varying physical phenomena inevitably creates artificial discontinuities and rigid boundary constraints.

To address these limitations, **Fuzzy Logic** extends classical Boolean operations into a multi-valued continuum. By replacing binary characteristic functions with continuous membership functions, fuzzy logic provides a rigorous mathematical framework for representing vagueness, uncertainty, and expert human reasoning.

Example 5.1 (Crisp vs. Fuzzy Boundary Constraint)**.** *Consider a temperature threshold defined for a thermal regulator system over* $X = [0, 50]\ ^\circ C$*:*

$$\textit{Crisp Set ("Hot"):}\quad \chi_{Hot}(x) = \begin{cases} 1, & x \geq 30^\circ C \\ 0, & x < 30^\circ C \end{cases}$$

$$\textit{Fuzzy Set ("Hot"):}\quad \mu_{Hot}(x) = \begin{cases} 0, & x \leq 20^\circ C \\ \dfrac{x - 20}{10}, & 20^\circ C < x < 30^\circ C \\ 1, & x \geq 30^\circ C \end{cases}$$

Under the crisp model, $29.9^\circ C$ *is treated as completely non-hot* ($\chi = 0$)*, whereas* $30.0^\circ C$ *is fully hot* ($\chi = 1$)*. The fuzzy membership function* $\mu_{Hot}(x)$ *smooths this transition across* $[20, 30]\ ^\circ C$*, preventing abrupt control actions.*

5.2 Fuzzy propositions and their truth

Definition 5.1: Fuzzy proposition

A fuzzy proposition $\widetilde{P}$ is a statement involving some concept *without clearly defined boundaries.* Linguistic statements that express subjective ideas, and that can be interpreted slightly differently by various individuals, typically involve fuzzy propositions; most natural language is fuzzy. The truth value assigned to $\widetilde{P}$ can be any value on the interval $[0, 1]$:

$$T:\ u \in U \longrightarrow [0, 1]. \tag{5.1}$$

Fuzzy propositions are assigned to *fuzzy sets.* If proposition $\widetilde{P}$ is assigned to fuzzy set $\widetilde{A}$, then the truth value of the proposition is given by

$$T(\widetilde{P}) = \mu_{\widetilde{A}}(x), \qquad 0 \leq \mu_{\widetilde{A}} \leq 1. \tag{5.2}$$

the degree of truth for the proposition $\widetilde{P} : x \in \widetilde{A}$ equals the membership grade of x in the fuzzy set $\widetilde{A}$.

5.3 Fuzzy logical connectives

The logical connectives of negation, disjunction, conjunction, and implication are defined for fuzzy logic for two simple propositions $\widetilde{P}$ (on fuzzy set $\widetilde{A}$) and $\widetilde{Q}$ (on fuzzy set $\widetilde{B}$) as follows.

Fuzzy connectives

$$\textit{Negation:} \quad T(\overline{\widetilde{P}}) = 1 - T(\widetilde{P}), \tag{5.3}$$

$$\textit{Disjunction:} \quad \widetilde{P} \vee \widetilde{Q} : x \text{ is } \widetilde{A} \text{ or } \widetilde{B}, \quad T(\widetilde{P} \vee \widetilde{Q}) = \max\big(T(\widetilde{P}), T(\widetilde{Q})\big), \tag{5.4}$$

$$\textit{Conjunction:} \quad \widetilde{P} \wedge \widetilde{Q} : x \text{ is } \widetilde{A} \text{ and } \widetilde{B}, \quad T(\widetilde{P} \wedge \widetilde{Q}) = \min\big(T(\widetilde{P}), T(\widetilde{Q})\big), \tag{5.5}$$

$$\textit{Implication (Zadeh, 1973):} \quad \widetilde{P} \implies \widetilde{Q} : x \text{ is } \widetilde{A}, \text{ then } x \text{ is } \widetilde{B},$$

$$T(\widetilde{P} \implies \widetilde{Q}) = T(\widetilde{P} \vee \widetilde{Q}) = \max\big(T(\widetilde{P}), T(\widetilde{Q})\big). \tag{5.6}$$

5.4 Fuzzy Implication

As in binary logic, the implication can be modelled in rule-based form: $\widetilde{P} \implies \widetilde{Q}$ is "IF x is $\widetilde{A}$, THEN y is $\widetilde{B}$", equivalent to the fuzzy relation

$$\widetilde{R} = (\widetilde{A} \times \widetilde{B}) \cup (\overline{\widetilde{A}} \times Y), \tag{5.7}$$

with membership function

$$\mu_{\widetilde{R}}(x, y) = \max\big[\min\big(\mu_{\widetilde{A}}(x),\, \mu_{\widetilde{B}}(y)\big),\; 1 - \mu_{\widetilde{A}}(x)\big]. \tag{5.8}$$

Types of Fuzzy Propositions:

- **Atomic Fuzzy Proposition:** A single statement of the form "x is A", where x is a linguistic variable and A is a linguistic value (fuzzy set).
- **Compound Fuzzy Proposition:** A combination of atomic propositions connected by *and*, *or*, and *not*.

Example 5.2.

When the conditional implication is of the compound form

$$\text{IF } x \text{ is } \widetilde{A}, \text{ THEN } y \text{ is } \widetilde{B}, \text{ ELSE } y \text{ is } \widetilde{C},$$

the equivalent fuzzy relation is $\widetilde{R} = (\widetilde{A} \times \widetilde{B}) \cup (\overline{\widetilde{A}} \times \widetilde{C})$, just as in (4.13), with membership function

$$\mu_{\widetilde{R}}(x, y) = \max\big[\min\big(\mu_{\widetilde{A}}(x),\, \mu_{\widetilde{B}}(y)\big),\, \min\big(1 - \mu_{\widetilde{A}}(x),\, \mu_{\widetilde{C}}(y)\big)\big]. \tag{5.9}$$

Example 5.3.

5.4.1 Other forms of the implication operation

There are other techniques for obtaining the fuzzy relation $\widetilde{R}$ based on IF $\widetilde{A}$, THEN $\widetilde{B}$, i.e. $\widetilde{R} = \widetilde{A} \implies \widetilde{B}$. These are known as *fuzzy implication operations*, and they are valid for all values of $x \in X$ and $y \in Y$. The following forms show different techniques for obtaining the membership function values of the fuzzy relation $\widetilde{R}$ defined on the Cartesian product space $X \times Y$:

Table 6: Fuzzy implication operations

Name	**Symbol**	**Membership Function Formula** $\mu_{\widetilde{R}}(x,y)$
Dienes-Rescher	Q_D	$\max\left[\mu_{\widetilde{B}}(y),\ 1-\mu_{\widetilde{A}}(x)\right]$
Lukasiewicz	Q_L	$\mu_{\widetilde{R}}(x,y)=\min\left\{1,\ \left[1-\mu_{\widetilde{A}}(x)+\mu_{\widetilde{B}}(y)\right]\right\}$
Zadeh	Q_Z	$\max\left[\min\left(\mu_{\widetilde{A}}(x),\,\mu_{\widetilde{B}}(y)\right),\ 1-\mu_{\widetilde{A}}(x)\right]$
Gödel	Q_G	$\begin{cases}1, & \mu_{\widetilde{A}}(x)\le\mu_{\widetilde{B}}(y),\\ \mu_{\widetilde{B}}(y), & \text{otherwise.}\end{cases}$
Mamdani Minimum	Q_{MM}	$\min\left[\mu_{\widetilde{A}}(x),\,\mu_{\widetilde{B}}(y)\right]$
Mamdani Product	Q_{MP}	$\mu_{\widetilde{A}}(x)\cdot\mu_{\widetilde{B}}(y)$

Lemma 5.1. *For all $(x,y)\in X\times Y$, the ordering relationship among Q_Z, Q_D, and Q_L is given by:*

$$\mu_{Q_Z}(x,y)\le\mu_{Q_D}(x,y)\le\mu_{Q_L}(x,y)$$

5.5 Fuzzy Composition

Fuzzy Composition is the primary operation used in fuzzy logic to combine a fuzzy set (an input or observation) with a fuzzy relation (a rule or mapping) to infer a new fuzzy set as the output. It forms the mathematical backbone of the *Compositional Rule of Inference* (Zadeh, 1973).

Mathematical Formulation

Let X be an input universe of discourse and Y be an output universe of discourse.

- Let $\widetilde{A}'$ be a fuzzy set defined on X with membership function $\mu_{\widetilde{A}'}(x)$.
- Let $\widetilde{R}$ be a fuzzy binary relation on $X\times Y$ with membership matrix $\mu_{\widetilde{R}}(x,y)$.

The composition of $\widetilde{A}'$ and $\widetilde{R}$, denoted by $\widetilde{B}'=\widetilde{A}'\circ\widetilde{R}$, yields a new fuzzy set $\widetilde{B}'$ defined on Y:

$$\widetilde{B}'=\widetilde{A}'\circ\widetilde{R} \tag{5.10}$$

Common Composition Operators

The membership degree $\mu_{\widetilde{B}'}(y)$ for each element $y\in Y$ is evaluated as follows:

A. **Max–Min Composition:** The most widely used composition method in fuzzy control (Mamdani systems):

$$\mu_{\widetilde{B}'}(y)=\max_{x\in X}\left\{\min\left[\mu_{\widetilde{A}'}(x),\,\mu_{\widetilde{R}}(x,y)\right]\right\}=\bigvee_{x\in X}\left[\mu_{\widetilde{A}'}(x)\wedge\mu_{\widetilde{R}}(x,y)\right] \tag{5.11}$$

B. **Max–Product Composition:** Also known as correlation-product composition, using algebraic multiplication instead of minimum:

$$\mu_{\widetilde{B}'}(y)=\max_{x\in X}\left\{\mu_{\widetilde{A}'}(x)\cdot\mu_{\widetilde{R}}(x,y)\right\}=\bigvee_{x\in X}\left[\mu_{\widetilde{A}'}(x)\cdot\mu_{\widetilde{R}}(x,y)\right] \tag{5.12}$$

C. **Sum–Product Composition:** Often used in adaptive neuro-fuzzy inference systems (ANFIS):

$$\mu_{\widetilde{B}'}(y) = \min\left[1, \sum_{x\in X} \mu_{\widetilde{A}'}(x) \cdot \mu_{\widetilde{R}}(x,y)\right] \tag{5.13}$$

Algebraic Properties of Fuzzy Composition

For fuzzy relations $\widetilde{R} \subseteq X \times Y$, $\widetilde{S} \subseteq Y \times Z$, and $\widetilde{T} \subseteq Z \times W$:

Table 7: Properties of compositional operators

Property	**Mathematical Formulation**
Associativity	$(\widetilde{R} \circ \widetilde{S}) \circ \widetilde{T} = \widetilde{R} \circ (\widetilde{S} \circ \widetilde{T})$
Distributivity over Union	$\widetilde{R} \circ (\widetilde{S} \cup \widetilde{T}) = (\widetilde{R} \circ \widetilde{S}) \cup (\widetilde{R} \circ \widetilde{T})$
Monotonicity	$\widetilde{S} \subseteq \widetilde{T} \implies (\widetilde{R} \circ \widetilde{S}) \subseteq (\widetilde{R} \circ \widetilde{T})$
Inverse Relation	$(\widetilde{R} \circ \widetilde{S})^{-1} = \widetilde{S}^{-1} \circ \widetilde{R}^{-1}$

Example 5.4.

5.5.1 Non-Existence of the Inverse in Fuzzy Composition

A fundamental difference between classical Boolean logic and fuzzy logic lies in the concept of reversibility (or inverse mapping).

In classical binary logic, applying modus ponens with the original input A and relation R perfectly recovers the exact original output B:

$$B = A \circ R. \tag{5.14}$$

In approximate reasoning, however, **an inverse does not exist in general**. If we compose the original antecedent set $\widetilde{A}$ with the fuzzy relation matrix $\widetilde{R}$, the resulting fuzzy set $\widetilde{B}' = \widetilde{A} \circ \widetilde{R}$ is generally **not equal** to the original consequent set $\widetilde{B}$:

$$\widetilde{B}' = \widetilde{A} \circ \widetilde{R} \neq \widetilde{B}. \tag{5.15}$$

Why Does the Inverse Fail?

The composition operator $\circ$ uses min (intersection) and max (union). These operators are non-injective (many-to-one mappings) and introduce **fuzziness propagation** or information loss. As a result, the inferred set $\widetilde{B}'$ tends to be more diffuse (has higher entropy / uncertainty) than the original set $\widetilde{B}$.

Example 5.5.

Example 5.6.

6 Linguistic Variables, Connectives, and Hedges

In classical mathematics, variables assume precise numerical values (e.g., $v = 50$ km/h). However, human reasoning and natural communication rely on conceptual patterns and qualitative descriptions (e.g., "the car is moving fast"). Fuzzy set theory bridges this gap through **linguistic variables**, providing a quantitative calculus for natural language.

6.1 Linguistic Variables: Conceptual & Formal Definitions

Definition 6.1 (Linguistic Variable)**.** *A **linguistic variable** is a variable whose values are words or sentences in a natural or artificial language rather than numbers. Each linguistic word (term) is modeled as a fuzzy set defined over a physical universe of discourse.*

Formally, a linguistic variable is characterized by a 4-tuple:

$$(X,\, T,\, U,\, M) \tag{6.1}$$

where:

- X: The name of the linguistic variable (e.g., *Speed*).
- T: The term-set, i.e., the collection of linguistic values/labels that X can assume (e.g., $T = \{\text{slow}, \text{medium}, \text{fast}\}$).
- U: The physical domain or universe of discourse (e.g., $U = [0, 120]$ km/h).
- M: The semantic rule mapping each term $t \in T$ to a fuzzy set on U defined by its membership function $\mu_t(u)$.

Example 6.1. *Let X = Age on the physical universe of discourse $U = [0, 100]$ years. The semantic mapping $M$for the primary terms"young" and "old" can be defined continuous-wise as:*

$$\mu_{young}(y) = \begin{cases} 1, & y \le 25, \\ \left[1 + \left(\dfrac{y-25}{5}\right)^2\right]^{-1}, & y > 25, \end{cases} \tag{6.2}$$

$$\mu_{old}(y) = 1 - \left[1 + \left(\frac{y-50}{5}\right)^2\right]^{-1}, \quad 50 \le y \le 100. \tag{6.3}$$

6.2 Composites: Linguistic Connectives

Composite expressions are formed by combining atomic linguistic terms using set-theoretic operators corresponding to natural language connectives (*AND*, *OR*, *NOT*):

For two terms α and β defined on U:

$$\text{Union } (OR): \quad \mu_{\alpha \text{ OR } \beta}(u) = \max\left(\mu_\alpha(u),\, \mu_\beta(u)\right) = \mu_\alpha(u) \vee \mu_\beta(u), \tag{6.4}$$

$$\text{Intersection } (AND): \quad \mu_{\alpha \text{ AND } \beta}(u) = \min\left(\mu_\alpha(u),\, \mu_\beta(u)\right) = \mu_\alpha(u) \wedge \mu_\beta(u), \tag{6.5}$$

$$\text{Complement } (NOT): \quad \mu_{\text{NOT } \alpha}(u) = 1 - \mu_\alpha(u). \tag{6.6}$$

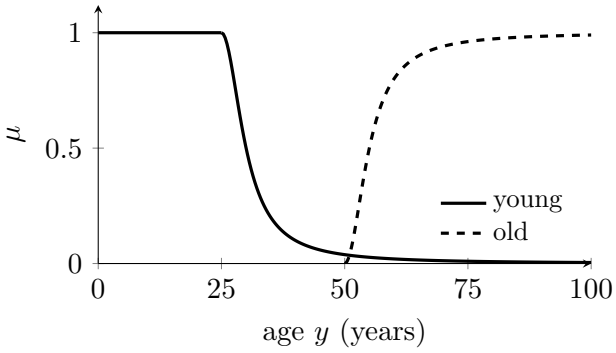


Figure 4: Fuzzy interpretations of the linguistic atoms "young" and "old" on the universe of ages

6.3 Linguistic Hedges

A **linguistic hedge** is an operator/modifier (adverb or adjective) that alters the membership function of a primary fuzzy set to represent intensifiers or diluators such as *very*, *slightly*, or *more or less*.

Mathematical Operators for Primary Hedges (Zadeh, 1972)

Let $\widetilde{A}$ be a fuzzy set defined on U with membership function $\mu_A(u)$.

Concentration (*very*): $$\mu_{\text{very } \widetilde{A}}(u) = \left[\mu_{\widetilde{A}}(u)\right]^2, \tag{6.7}$$

Extreme Concentration (*very very*): $$\mu_{\text{very very } \widetilde{A}}(u) = \left[\mu_{\widetilde{A}}(u)\right]^4, \tag{6.8}$$

Plus: $$\mu_{\text{plus } \widetilde{A}}(u) = \left[\mu_{\widetilde{A}}(u)\right]^{1.25}, \tag{6.9}$$

Dilation (*slightly / more or less*): $$\mu_{\text{more or less } \widetilde{A}}(u) = \left[\mu_{\widetilde{A}}(u)\right]^{0.5}, \tag{6.10}$$

Minus: $$\mu_{\text{minus } \widetilde{A}}(u) = \left[\mu_{\widetilde{A}}(u)\right]^{0.75}, \tag{6.11}$$

Intensification: $$\mu_{\text{INT}(\widetilde{A})}(u) = \begin{cases} 2\left[\mu_{\widetilde{A}}(u)\right]^2, & 0 \le \mu_{\widetilde{A}}(u) \le 0.5, \\ 1 - 2\left[1 - \mu_{\widetilde{A}}(u)\right]^2, & 0.5 < \mu_{\widetilde{A}}(u) \le 1. \end{cases} \tag{6.12}$$

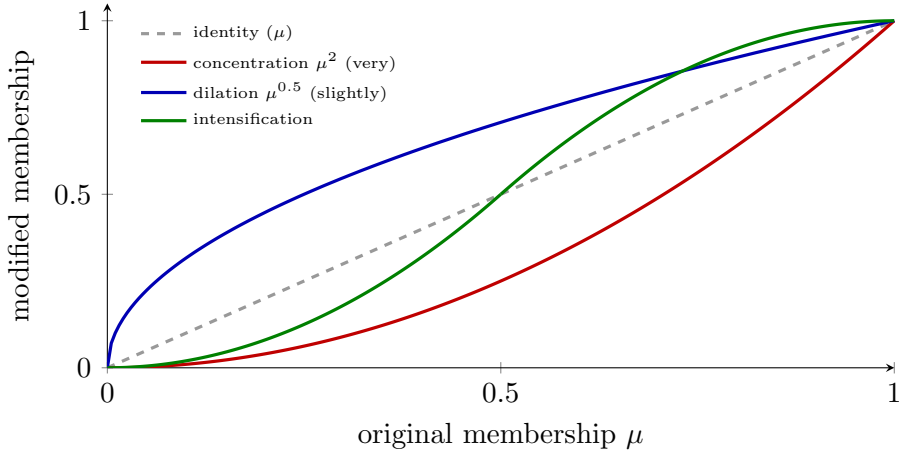


Figure 5: Fuzzy concentration, dilation, and intensification of a fuzzy set

Physical Effect of Hedges

- **Concentration (μ^2):** Reduces the membership values of elements that are only "partially" in the set. For $0 < \mu < 1$, $\mu^2 < \mu$. (e.g., $\mu = 0.9 \to 0.81$, but $\mu = 0.1 \to 0.01$).
- **Dilation ($\mu^{0.5}$):** Stretches the fuzzy set by increasing membership values across intermediate regions. (e.g., $\mu = 0.81 \to 0.9$, and $\mu = 0.01 \to 0.1$).
- **Intensification (INT):** Increases contrast by making the slope of the membership function steeper around the crossover point ($\mu = 0.5$). It boosts values where $\mu > 0.5$ and suppresses values where $\mu < 0.5$.

6.4 Operator Precedence and Composite Evaluation

To systematically evaluate complex natural language expressions (such as **"not very small AND not very, very large"**), operations must follow a standard precedence order:

Table 8: Precedence for linguistic hedges and set operations (Zadeh, 1973)

Order	**Operation**	**Associativity / Direction**
1st	Hedges (*very*, *more or less*) & *NOT*	Right-to-left evaluation
2nd	*AND* (Intersection)	Left-to-right evaluation
3rd	*OR* (Union)	Left-to-right evaluation

Parentheses override default precedence. Ambiguous nested hedges evaluate **right-to-left**:

$$\text{"plus very minus small"} \;\equiv\; \text{plus}\Big(\text{very}\big(\text{minus}(\text{small})\big)\Big).$$

Example 6.2.

7 Fuzzy Rule-Based Systems

In artificial intelligence and machine intelligence, various paradigms exist for formalizing human knowledge. The most widespread representation expresses empirical knowledge as natural language conditional statements:

$$\text{IF premise (antecedent), THEN conclusion (consequent)} \tag{7.1}$$

This form is commonly referred to as the IF–THEN rule-based form; this form is generally referred to as the deductive form. It typically expresses an inference such that if we know a fact (premise, hypothesis, antecedent), then we can infer, or derive, another fact called a conclusion (consequent).

7.1 Shallow vs. Deep Forms of Knowledge

Knowledge used by an expert system comes in two very different forms.

- **Shallow knowledge** is "rule-of-thumb" knowledge: the kind of empirical, heuristic reasoning people express in everyday language. It links what we *observe* directly to what we *conclude*, usually as simple deductive rules of the form "if this, then that." It is easy to write down and easy to reason with.
- **Deep knowledge** is the knowledge of *why* things behave as they do: the underlying physical structure, how the parts interact and function, intuitive insight, and the behaviour of complex objects. This kind of knowledge is hard to capture in a handful of plain-language rules.

Example 7.1.

Table 9: Shallow vs. deep knowledge

Example	**Shallow knowledge (rule)**	**Deep knowledge (model)**
Medical diagnosis	"If the patient has a fever and a cough, then suspect influenza." Learned from experience; needs no theory; maps symptoms straight to a conclusion.	A model of how the virus infects the respiratory tract, how the immune response produces fever, and how the disease evolves over time. Explains *why* the symptoms occur and predicts what happens next — but cannot be written as one "if…, then…" line.
Process control	"If the temperature is high, then open the cooling valve more." A typical fuzzy control rule.	The energy-balance (thermodynamic) model of the plant: how heat enters, is stored, and is removed. Predicts the effect of *any* valve setting, not just the ones seen before.
Everyday reasoning	"If the car will not start and the headlights are dim, then the battery is flat."	Understanding the electrical circuit, the battery's chemistry, and the role of the alternator — which also allows diagnosing a fault never seen before.

Fuzzy rule-based systems bridge the gap between shallow and deep knowledge. They retain the rule form of the shallow layer — antecedents and consequents stated as **linguistic variables** — yet fuzzy sets and logical connectives turn the whole rule base into an explicit mathematical model: one that interpolates smoothly between rules, predicts for inputs covered by no single memorised rule, and remains open to analysis and design.

7.2 The canonical form of fuzzy rule bases

A fuzzy rule-based description of a complex system operates as a set of output restrictions conditioned on input states. Complex compound rule structures can be mathematically decomposed into simple canonical forms. The canonical base is displayed in Table 10.

Table 10: Canonical form for a fuzzy rule-based system

Rule Index	**Canonical IF–THEN Form**
Rule 1	IF condition C^1, THEN restriction R^1
Rule 2	IF condition C^2, THEN restriction R^2
$\vdots$	$\vdots$
Rule r	IF condition C^r, THEN restriction R^r

Here, the conditions $C^1, C^2, \ldots, C^r$ characterise the input state space, while the restrictions $R^1, R^2, \ldots, R^r$ define the output fuzzy relations and action boundaries. Written as a single-valued output rule — the *canonical singleton-relation form* — each line reads

$$\text{IF } x \text{ is } \widetilde{A}^s \text{ THEN } y \text{ is } \widetilde{B}^s, \qquad s = 1, \ldots, r, \tag{7.2}$$

where $\widetilde{A}^s$ is a fuzzy set on the input universe X and $\widetilde{B}^s$ is a fuzzy set on the output universe Y. The pivotal claim is that *every* rule base of practical interest can be brought to the form (7.2): whatever compound structure the antecedents and consequents originally possess, it reduces to a single antecedent set and a single consequent set. The following section makes that reduction explicit.

7.2.1 Decomposition techniques for compound antecedents

Compound rules featuring multiple antecedents connected by linguistic operators (e.g. "AND," "OR") are decomposed into the standard canonical form using fuzzy set operations. We treat the conjunctive and disjunctive cases in turn, then combine them.

- **Multiple conjunctive antecedents**

  A compound rule with antecedents connected by logical conjunctions (AND) has the surface form

  $$\text{IF } x_1 \text{ is } \widetilde{A}^1 \text{ AND } x_2 \text{ is } \widetilde{A}^2 \ \ldots \ \text{AND } x_L \text{ is } \widetilde{A}^L \ \text{ THEN } y \text{ is } B^s. \tag{7.3}$$

  It is reduced to the canonical statement

  $$\text{IF } \widetilde{A}^s \text{ THEN } \widetilde{B}^s,$$

  where $\widetilde{A}^s$ represents the composite fuzzy intersection set

  $$\widetilde{A}^s = \widetilde{A}^1 \cap \widetilde{A}^2 \cap \cdots \cap \widetilde{A}^L. \tag{7.4}$$

  Because each antecedent predicate x_ℓ is $\widetilde{A}^\ell$ is defined over the *same* input vector $\mathbf{x} = (x_1, \ldots, x_L)$, membership in the composite set (7.4) is evaluated pointwise. Under standard fuzzy logic the composite membership grade is computed as follows:

  $$\mu_{\widetilde{A}^s}(x) = \min\big[\, \mu_{\widetilde{A}^1}(x),\ \mu_{\widetilde{A}^2}(x),\ \ldots,\ \mu_{\widetilde{A}^L}(x) \big]\,. \tag{7.5}$$

The minimum returns the *weakest* degree among the conjuncts; the rule fires only as strongly as its least-satisfied precondition. This is the formal content of the familiar engineering intuition that a conjunction of conditions is only as good as its weakest link.

- **Multiple disjunctive antecedents**

  A compound rule with antecedents connected by logical disjunctions (OR) has the surface form

  $$\text{IF } x_1 \text{ is } \widetilde{A}^1 \text{ OR } x_2 \text{ is } \widetilde{A}^2 \ \dots \ \text{OR } x_L \text{ is } \widetilde{A}^L \ \text{ THEN } y \text{ is } B^s. \tag{7.6}$$

  By the dual decomposition it reduces to the canonical statement IF $\widetilde{A}^s$ THEN $\widetilde{B}^s$, where now $\widetilde{A}^s$ is the composite fuzzy union set,

  $$\widetilde{A}^s = \widetilde{A}^1 \cup \widetilde{A}^2 \cup \cdots \cup \widetilde{A}^L, \tag{7.7}$$

  and the composite membership grade is computed as follows:

  $$\mu_{\widetilde{A}^s}(x) = \max\big[\, \mu_{\widetilde{A}^1}(x),\ \mu_{\widetilde{A}^2}(x),\ \dots,\ \mu_{\widetilde{A}^L}(x)\big]\,. \tag{7.8}$$

  The maximum returns the *strongest* degree among the disjuncts: a disjunction is satisfied to the extent that its best-satisfied alternative is satisfied.

- **Mixed and nested connectives**

  Real rule bases routinely interleave the two operators and introduce grouping, for example

  $$\text{IF } \big(x_1 \text{ is } \widetilde{A}^1 \text{ AND } x_2 \text{ is } \widetilde{A}^2\big) \text{ OR } \big(x_3 \text{ is } \widetilde{A}^3\big) \ \text{ THEN } y \text{ is } \widetilde{B}^s. \tag{7.9}$$

  Such a rule is decomposed by evaluating the expression tree bottom-up, replacing every AND-node by min and every OR-node by max:

  $$\mu_{\widetilde{A}^s}(x) = \max\Big[\, \min\big(\mu_{A^1}(x),\, \mu_{\widetilde{A}^2}(x)\big),\ \mu_{\widetilde{A}^3}(x)\Big]. \tag{7.10}$$

Example 7.2 (Conjunctive decomposition and COG)**.** *Let a temperature controller use a single rule*

IF temp is high AND *pres. is* high *THEN valve is* slightly open.

Suppose at input $x^* = (t^*, p^*)$ the two antecedent memberships are

$$\mu_{high}(t^*) = 0.80, \qquad \mu_{high}(p^*) = 0.60.$$

By the conjunctive decomposition (7.5), the composite antecedent degree is

$$\mu_{\widetilde{A}^s}(x^*) = \min(0.80,\, 0.60) = 0.60,$$

so the rule fires with strength $\omega = 0.60$. Under Mamdani implication the consequent set "slightly open" is truncated to height 0.60; defuzzifying by the centroid returns the crisp valve opening. Had the same rule used OR *instead of* AND*, the disjunctive decomposition (7.8) would give a firing strength of $\max(0.80, 0.60) = 0.80$, a materially larger action — illustrating how the logical connective, made explicit by the decomposition, changes the behaviour of the system.*

Example 7.3.

Example 7.4. *Consider the rule*

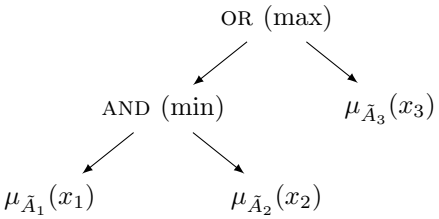
$$\textbf{\textit{IF}}\ x_1\ \textit{is}\ \tilde{A}_1\ \textbf{\texttt{and}}\ x_2\ \textit{is}\ \tilde{A}_2\ \textbf{\texttt{or}}\ x_3\ \textit{is}\ \tilde{A}_3\quad \textbf{\textit{THEN}}\ y\ \textit{is}\ \tilde{B}_s. \tag{8.9}$$

Such a rule is decomposed by evaluating the expression tree bottom-up, replacing every **`and`***-node by* min *and every* **`or`***-node by* max*. Because the connective* **`and`** *binds more tightly than* **`or`***, the antecedent of (8.9) parses as*

$$\big(x_1\ \textit{is}\ \tilde{A}_1\ \wedge\ x_2\ \textit{is}\ \tilde{A}_2\big)\ \vee\ \big(x_3\ \textit{is}\ \tilde{A}_3\big),$$

so the expression tree has an **`or`***-node at the root over an* **`and`***-node and a leaf. Evaluating bottom-up gives the firing strength of the whole antecedent:*

$$\mu_{\tilde{A}_s}(x)\ =\ \max\Big[\ \min\big(\mu_{\tilde{A}_1}(x_1),\ \mu_{\tilde{A}_2}(x_2)\big),\ \mu_{\tilde{A}_3}(x_3)\ \Big]. \tag{7.11}$$



Take the following membership grades for a given input vector (x_1, x_2, x_3)*:*

Term	*Grade*
$\mu_{\tilde{A}_1}(x_1)$	0.40
$\mu_{\tilde{A}_2}(x_2)$	0.60
$\mu_{\tilde{A}_3}(x_3)$	0.80

Step 1 — the **`and`***-node:* $\min(0.40,\ 0.60) = 0.40$.
Step 2 — the **`or`***-node:* $\max(0.40,\ 0.80) = 0.80$*. Hence, from* (7.11)*,*

$$\mu_{\tilde{A}_s}(x) = 0.80.$$

This value is the antecedent's overall degree of satisfaction; it then fires the consequent in the usual way. Under Mamdani inference,

$$\mu_{\tilde{B}'_s}(y)\ =\ \min\big(0.80,\ \mu_{\tilde{B}_s}(y)\big), \tag{7.12}$$

i.e. the consequent fuzzy set $\tilde{B}_s$ *is clipped at* 0.80*.*

Why the grouping matters.

The parenthesisation is not cosmetic. Regrouping the same expression as $\tilde{A}_1 \wedge (\tilde{A}_2 \vee \tilde{A}_3)$ *gives, for the grades above,*

$$\min\big(0.40,\ \max(0.60,\ 0.80)\big)\ =\ \min(0.40,\ 0.80)\ =\ 0.40,$$

*a different answer (*0.40 *versus* 0.80*) from the same three inputs. The two groupings agree*

only in special cases, so a rule base that interleaves the connectives must fix its precedence convention — or insert explicit parentheses — before the bottom-up evaluation is well defined. Most texts adopt the classical precedence: `and` *first, then* `or`*.*

7.2.2 Rule Aggregation Methods

Most rule-based systems rely on multiple rules. Combining the individual conclusions from these rules to reach a single final outcome is called **rule aggregation**. When choosing an aggregation strategy, two main extreme approaches exist:

- **Conjunctive Aggregation (Joint Satisfaction):** When all rules must be satisfied simultaneously, rules are connected by AND operators. The aggregated output is determined by the intersection of all individual rule consequents $y^1, y^2, \ldots, y^r$:

$$y = y^1 \cap y^2 \cap \cdots \cap y^r \tag{7.13}$$

$$\mu_y(y) = \min\left(\mu_{y^1}(y), \mu_{y^2}(y), \ldots, \mu_{y^r}(y)\right), \quad \forall y \in Y \tag{7.14}$$

- **Disjunctive Aggregation (At Least One Satisfaction):** When the system requires satisfaction of at least one rule, rules are connected by OR operators. The aggregated output is determined by the union of all individual rule consequents:

$$y = y^1 \cup y^2 \cup \cdots \cup y^r \tag{7.15}$$

$$\mu_y(y) = \max\left(\mu_{y^1}(y), \mu_{y^2}(y), \ldots, \mu_{y^r}(y)\right), \quad \forall y \in Y \tag{7.16}$$

Example 7.5.

7.3 Fuzzy Inference System

A **Fuzzy Inference System (FIS)** is a computing framework that reasons with *linguistic rules* instead of exact equations — it turns human-style statements such as *"IF temperature is high AND humidity is low THEN fan speed is fast"* into precise numeric outputs. Crisp inputs are fuzzified into membership degrees, a rule base is fired in parallel to yield per-rule firing strengths, the fired rules' outputs are combined, and the result is defuzzified back into one crisp number. The three main flavors — **Mamdani** (fuzzy-set consequents, most interpretable), **TSK** (function consequents, fastest and most learnable), and **Tsukamoto** (monotonic consequents, historical compromise) — all follow this same skeleton, but they diverge on **one structural decision: what sits on the right-hand side of each rule**. Interpretability, speed, precision, and learnability all flow from that choice.

7.3.1 Mamdani Fuzzy Inference System

Proposed by Ebrahim Mamdani in 1975, the **Mamdani Fuzzy Inference System (FIS)** was designed to control a steam engine and boiler combination by synthesizing human expert knowledge into fuzzy IF-THEN rules. It maps a crisp numerical input vector $\mathbf{x}_0 \in \mathbb{R}^n$ to a crisp scalar output $z^* \in \mathbb{R}$ using fuzzy set representations in both antecedents and consequents.

Architecture and Pipeline: A standard Mamdani FIS operates through four primary computational stages:

1. **Fuzzification:** Crisp inputs are mapped to degrees of membership within input fuzzy sets.
2. **Rule Evaluation (Implication):** Firing strengths are computed via T-norms, and rule consequents are shaped.
3. **Aggregation:** Individual modified consequent fuzzy sets are combined via S-norms into a single aggregated fuzzy output set.
4. **Defuzzification:** The aggregated fuzzy output distribution is resolved into a single scalar value.

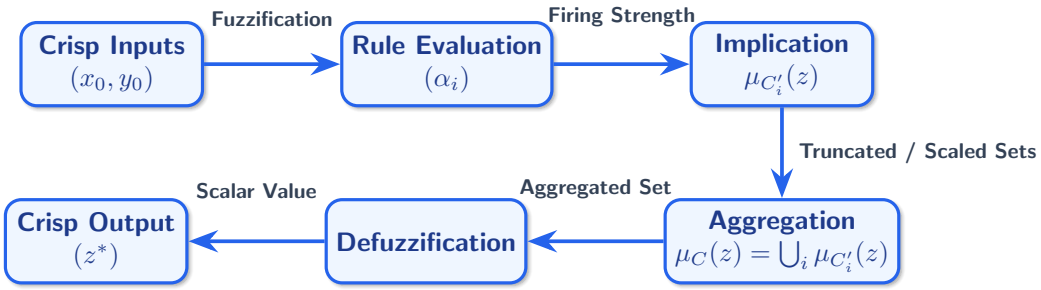


Figure 6: Block flow of the Mamdani Fuzzy Inference Pipeline.

Mathematical Formulation: Consider a fuzzy rule base consisting of N rules of the form:

$$R_i : \text{IF } x \text{ is } A_i \text{ AND } y \text{ is } B_i \text{ THEN } z \text{ is } C_i, \quad i = 1, 2, \ldots, N \tag{7.17}$$

- **Fuzzification**

  Given crisp inputs $x = x_0$ and $y = y_0$, the membership degrees are evaluated as:

$$\mu_{A_i}(x_0) \in [0, 1], \quad \mu_{B_i}(y_0) \in [0, 1] \tag{7.18}$$

- **Rule Evaluation and Implication**

  For each rule R_i, we calculate its **firing strength** α_i (how strongly the rule applies) by combining the membership values of the inputs using an AND operator—most commonly the minimum or the product:

$$\alpha_i = \min\left(\mu_{A_i}(x_0), \mu_{B_i}(y_0)\right) \tag{7.19}$$

  Once α_i is determined, it is used to modify the output fuzzy set $\mu_{C_i}(z)$ to produce the rule's conclusion $\mu_{C_i'}(z)$. This is typically done in one of two ways:

  – **Mamdani Minimum (Truncation):** Cuts off (clips) the top of the output membership curve at height α_i:

$$\mu_{C_i'}(z) = \min\left(\alpha_i, \mu_{C_i}(z)\right) \tag{7.20}$$

– **Product Implication (Scaling):** Scales (squashes) the entire output membership curve proportionally by α_i:

$$\mu_{C_i'}(z) = \alpha_i \cdot \mu_{C_i}(z) \tag{7.21}$$

- **Aggregation**

  The output fuzzy sets $\mu_{C_i'}(z)$ from all active rules are combined into a single aggregated fuzzy set $\mu_C(z)$ using an S-norm operator (typically maximum or bounded sum):

$$\mu_C(z) = \bigcup_{i=1}^{N} \mu_{C_i'}(z) = \max_{i \in \{1,\dots,N\}} \left(\mu_{C_i'}(z)\right) \tag{7.22}$$

- **Defuzzification Methods**

  To convert the aggregated fuzzy set distribution $\mu_C(z)$ back into a crisp actionable output z^*, common techniques include:

  – **Centroid / Center of Gravity (COG):**

$$z^* = \frac{\int_{-\infty}^{\infty} z \cdot \mu_C(z)\, dz}{\int_{-\infty}^{\infty} \mu_C(z)\, dz} \tag{7.23}$$

  – **Center of Sums (COS):**

$$z^* = \frac{\sum_{i=1}^{N} z_i \cdot \text{Area}(C_i')}{\sum_{i=1}^{N} \text{Area}(C_i')} \tag{7.24}$$

  – **Mean of Maxima (MOM):**

$$z^* = \frac{\int_M z\, dz}{\int_M dz}, \quad \text{where } M = \left\{ z \middle| \mu_C(z) = \sup_{z'} \mu_C(z') \right\} \tag{7.25}$$

Example 7.6.

Example 7.7. *A restaurant wishes to automate its tipping recommendation system using a Mamdani Fuzzy Inference System (FIS). The system evaluates two crisp numerical inputs measured on a scale from* 0 *to* 10*:*

- ***Food Quality*** *(*$x \in [0, 10]$*):* *Evaluated at* $x_0 = 7.5$
- ***Service Quality*** *(*$y \in [0, 10]$*):* *Evaluated at* $y_0 = 4.0$

The objective is to compute the final crisp ***Tip Percentage*** *(*$z \in [0, 20]\%$*) by executing the complete 5-stage Mamdani inference pipeline under the definitions below.*

The input and output spaces are defined by the following membership functions:

(a) Food Quality Sets (x):

$$\mu_{Del}(x) = \begin{cases} \dfrac{x-5}{3}, & 5 \le x \le 8 \\ \dfrac{10-x}{2}, & 8 < x \le 10 \\ 0, & \textit{otherwise} \end{cases} \qquad \textit{and} \qquad \mu_{NOT\ Del}(x) = 1 - \mu_{Del}(x) \tag{7.26}$$

(b) Service Quality Sets (y):

$$\mu_{Poor}(y) = \begin{cases} 1 - \dfrac{y}{5}, & 0 \le y \le 5 \\ 0, & y > 5 \end{cases} \qquad \textit{and} \qquad \mu_{Good}(y) = \begin{cases} \dfrac{y}{5}, & 0 \le y \le 5 \\ 1, & y > 5 \end{cases} \tag{7.27}$$

(c) Output Tip Percentage Sets (z):

$$\mu_{Low}(z) = \begin{cases} 1 - \dfrac{z}{10}, & 0 \le z \le 10 \\ 0, & z > 10 \end{cases} \qquad \textit{and} \qquad \mu_{High}(z) = \begin{cases} \dfrac{z-10}{10}, & 10 \le z \le 20 \\ 0, & z < 10 \end{cases} \tag{7.28}$$

The decision rules governing the system are:

Rule 1 (R_1): *IF* Food is Delicious *OR* Service is Good *THEN* Tip is High

Rule 2 (R_2): *IF* Food is NOT Delicious *AND* Service is Poor *THEN* Tip is Low

Using standard Mamdani operators

(i) Compute the membership values for inputs $x_0 = 7.5$ and $y_0 = 4.0$.

(ii) Calculate the firing strengths α_1 and α_2 of rules R_1 and R_2.

(iii) Formulate the aggregated output membership function $\mu_C(z) = \max\left(\mu_{C_1'}(z), \mu_{C_2'}(z)\right)$.

(iv) Calculate the exact defuzzified crisp tip percentage $z^* = \frac{\int z\mu_C(z)dz}{\int \mu_C(z)dz}$.

7.3.2 Sugeno (TSK) Fuzzy Model

The Takagi-Sugeno-Kang (TSK) fuzzy model—introduced by Michio Sugeno, Tomohiro Takagi, and Michio Kang in 1985—is a widely used fuzzy inference framework in system modeling, adaptive control, and machine learning. Unlike the traditional Mamdani fuzzy inference system, which uses fuzzy sets for rule outputs (e.g., THEN Tip is High), the Sugeno model uses crisp mathematical functions of the inputs in its conclusions (e.g., THEN $y = 0.5x_1 + 1.2x_2 + 2$).

System Architecture & Rule Formulation: For a multi-input single-output (MISO) system with input vector $\mathbf{x} = [x_1, x_2, \ldots, x_n]^T \in \mathbb{R}^n$, the i-th fuzzy rule R_i $(i = 1, \ldots, N)$ is formulated as:

$$R_i : \text{IF } x_1 \text{ is } A_{i1} \text{ AND } x_2 \text{ is } A_{i2} \text{ AND } \ldots \text{ AND } x_n \text{ is } A_{in} \quad \text{THEN } y_i = f_i(x_1, x_2, \ldots, x_n) \tag{7.29}$$

The mathematical structure of the consequent function $f_i(\mathbf{x})$ defines the order of the model:

- **Zero-Order TSK Model:** $f_i(\mathbf{x}) = c_i$ (Constant output / Piecewise step function approximation)
- **First-Order TSK Model:** $f_i(\mathbf{x}) = p_{i0} + p_{i1}x_1 + p_{i2}x_2 + \cdots + p_{in}x_n = p_{i0} + \sum_{j=1}^{n} p_{ij}x_j$ (Affine linear combination)

Step-by-Step Mathematical Evaluation Pipeline: Given a specific crisp input vector $\mathbf{x}_0 = [x_{10}, x_{20}, \ldots, x_{n0}]^T$, the output of the TSK system is computed through four deterministic steps:

1. **Antecedent Membership Evaluation (Fuzzification):** Evaluate the degree of membership $\mu_{A_{ij}}(x_{j0}) \in [0, 1]$ for each input variable across all active fuzzy sets.

2. **Firing Strength Calculation:** Compute the activation weight w_i for rule R_i using a T-norm operator (typically minimum or algebraic product):

$$w_i = \prod_{j=1}^{n} \mu_{A_{ij}}(x_{j0}) \qquad \text{or} \qquad w_i = \min_{j \in \{1,\ldots,n\}} \left\{ \mu_{A_{ij}}(x_{j0}) \right\} \tag{7.30}$$

3. **Consequent Output Evaluation:** Substitute input values directly into the rule's consequent equation to derive the crisp individual rule prediction y_i:

$$y_i = p_{i0} + p_{i1}x_{10} + p_{i2}x_{20} + \cdots + p_{in}x_{n0} \tag{7.31}$$

4. **Aggregation via Normalized Weighted Average** Compute the global system output y^* as the weighted sum of individual rule consequents:

$$y^* = \frac{\sum_{i=1}^{N} w_i y_i}{\sum_{i=1}^{N} w_i} = \sum_{i=1}^{N} \bar{w}_i y_i \qquad \text{where} \qquad \bar{w}_i = \frac{w_i}{\sum_{k=1}^{N} w_k} \tag{7.32}$$

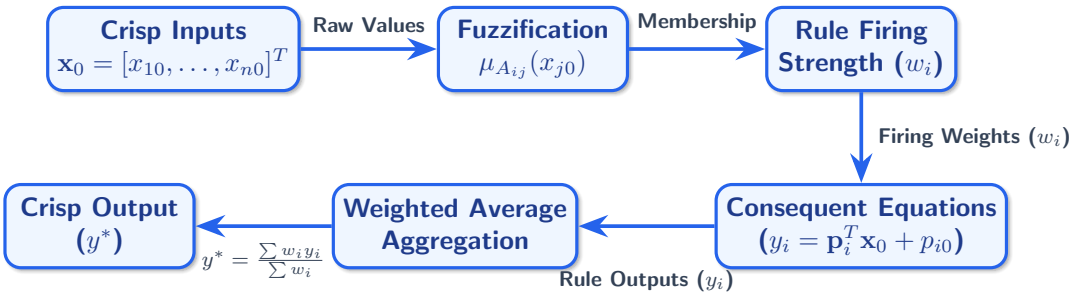


Figure 7: Block flow pipeline of the Takagi-Sugeno-Kang (TSK) Fuzzy Inference System.

Example 7.8. *Two rules, two inputs, one output. Let the crisp inputs be*

$$x = 4, \qquad y = 2.$$

Consider the rules,

$$\textit{Rule 1:} \quad \textit{IF } x \textit{ is } A_1 \textit{ and } y \textit{ is } B_1 \quad \textit{THEN } z_1 = 3x + 2y + 4,$$
$$\textit{Rule 2:} \quad \textit{IF } x \textit{ is } A_2 \textit{ and } y \textit{ is } B_2 \quad \textit{THEN } z_2 = 2x + 5y + 12.$$

with fuzzified input grades,

Rule	$\mu_{A_i}(x)$	$\mu_{B_i}(y)$
1	0.6	0.8
2	0.5	0.4

(a) Compute the firing strength w_i of each rule.

(b) Evaluate each rule's crisp output z_i.

(c) Compute the overall output z^* by the Sugeno weighted average.

Integration in Adaptive Neuro-Fuzzy Inference Systems (ANFIS)

The **first-order TSK** system forms the architectural foundation of **ANFIS (Adaptive Neuro-Fuzzy Inference System)** (Jang, 1993). ANFIS embeds the TSK architecture into a five-layer feedforward network: Layer 1 fuzzifies each input with adaptive membership functions, Layer 2 computes each rule's firing strength, Layer 3 normalizes the firing strengths, Layer 4 applies the linear consequents $z_i = p_{i0} + p_{i1}x_1 + \cdots + p_{in}x_n$, and Layer 5 sums to produce the overall system output y^*. Because the overall system output y^* is **linear in the consequent parameters** $(p_{i0}, p_{i1}, \dots)$, the parameters can be efficiently tuned using a **hybrid learning scheme** that splits the parameter space into two subsets, each handled by the method suited to it:

- **Forward pass (recursive least-squares estimation):** with the premise parameters held fixed, input signals propagate forward and the globally optimal linear consequent parameters (p_{ij}) are solved in a single pass.
- **Backward pass (gradient descent):** the error signals propagate backward, and the nonlinear antecedent premise parameters (e.g., Gaussian membership centers and standard deviations) are updated by gradient descent.

This division of labor converges far faster than pure gradient descent: least-squares estimation exploits the linearity of the consequents (a global optimum in that subspace), while gradient descent flexibly tunes the nonlinear membership functions.

7.3.3 Tsukamoto Fuzzy Model

The **Tsukamoto Fuzzy Model** (Tsukamoto, 1979) avoids the computational burden of traditional Mamdani defuzzification by enforcing a strict structural constraint: the membership functions of all consequent fuzzy sets must be **strictly monotonic**. Consequently, each rule produces a crisp output value via direct inverse mapping.

Mathematical Formulation: Consider a system with N rules of the form:

$$R_i : \text{IF } x_1 \text{ is } A_{i1} \text{ AND } x_2 \text{ is } A_{i2} \text{ AND } \dots \text{ AND } x_n \text{ is } A_{in} \text{ THEN } y \text{ is } C_i \tag{7.33}$$

where $\mu_{C_i}(y)$ is strictly monotonic over $y \in [y_{\min}, y_{\max}]$.

Inference Pipeline: Given an input vector $\mathbf{x}_0 = [x_{10}, x_{20}, \dots, x_{n0}]^T$:

1. **Fuzzification:** Evaluate the degree of membership for each crisp input component across the antecedent fuzzy sets:

$$\mu_{A_{ij}}(x_{j0}) \in [0,1] \quad \text{for } j = 1, 2, \dots, n \text{ and } i = 1, 2, \dots, N \tag{7.34}$$

2. **Firing Strength (α_i):** Compute the activation weight of rule R_i using a T-norm operator:

$$\alpha_i = \min_{j\in\{1,\dots,n\}} \left\{\mu_{A_{ij}}(x_{j0})\right\} \quad \text{or} \quad \alpha_i = \prod_{j=1}^{n} \mu_{A_{ij}}(x_{j0}) \tag{7.35}$$

3. **Crisp Rule Output (y_i):** Determine the individual output for rule R_i by evaluating the inverse of its strictly monotonic consequent membership function at level α_i:

$$y_i = \mu_{C_i}^{-1}(\alpha_i) \tag{7.36}$$

4. **Overall Defuzzified Output (y^*):** Calculate the final system output as the weighted average of the individual crisp rule outputs:

$$y^* = \frac{\sum_{i=1}^{N} \alpha_i y_i}{\sum_{i=1}^{N} \alpha_i} = \frac{\sum_{i=1}^{N} \alpha_i \mu_{C_i}^{-1}(\alpha_i)}{\sum_{i=1}^{N} \alpha_i} \tag{7.37}$$

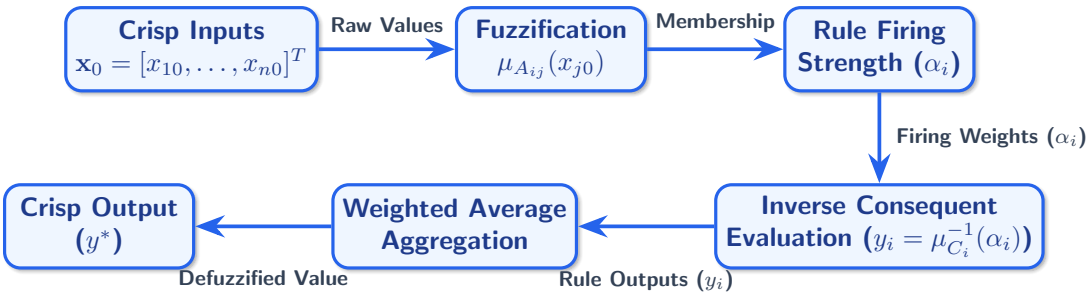


Figure 8: Block flow pipeline of the Tsukamoto Fuzzy Inference System.

Example 7.9. *A restaurant automates its tip recommendation using a Tsukamoto Fuzzy Inference System (FIS). The system evaluates two crisp inputs measured on a scale from* 0 *to* 10*:*

- ***Food Quality*** *($x_1 \in [0, 10]$):* *Evaluated at $x_1 = 7.5$*
- ***Service Quality*** *($x_2 \in [0, 10]$):* *Evaluated at $x_2 = 4.0$*

Compute the recommended ***Tip Percentage*** *$y \in [0, 20]\%$ by executing the Tsukamoto inference pipeline under the information specified below.*

- ***Food Quality*** *(x_1):*

$$\mu_{Del}(x_1) = \begin{cases} \dfrac{x_1 - 5}{3}, & 5 \le x_1 \le 8 \\ \dfrac{10 - x_1}{2}, & 8 < x_1 \le 10 \end{cases} \qquad \textit{and} \qquad \mu_{NOT\ Del}(x_1) = 1 - \mu_{Del}(x_1) \tag{7.38}$$

- ***Service Quality*** *(x_2):*

$$\mu_{Poor}(x_2) = 1 - \frac{x_2}{5}, \quad 0 \le x_2 \le 5 \qquad \textit{and} \qquad \mu_{Good}(x_2) = \frac{x_2}{5}, \quad 0 \le x_2 \le 5 \tag{7.39}$$

- *The output fuzzy sets are defined by strictly monotonic membership functions over* $y \in [0, 20]$*:*

$$\mu_{High}(y) = \frac{y}{20} \quad \Longrightarrow \quad y_1 = 20 \cdot \alpha_1 \tag{7.40}$$

$$\mu_{Low}(y) = \frac{20 - y}{20} \quad \Longrightarrow \quad y_2 = 20(1 - \alpha_2) \tag{7.41}$$

- *The decision logic is governed by two Tsukamoto-style rules:*

$$\textbf{Rule 1 } (\mathbf{R_1}): \quad \textbf{\textit{IF}} \textit{ Food is Delicious } \textbf{\textit{OR}} \textit{ Service is Good } \textbf{\textit{THEN}} \textit{ Tip is High} \tag{7.42}$$

$$\textbf{Rule 2 } (\mathbf{R_2}): \quad \textbf{\textit{IF}} \textit{ Food is NOT Delicious } \textbf{\textit{AND}} \textit{ Service is Poor } \textbf{\textit{THEN}} \textit{ Tip is Low} \tag{7.43}$$

(i) Evaluate the membership degrees $\mu_{Del}(7.5)$*,* $\mu_{NOT\ Del}(7.5)$*,* $\mu_{Good}(4.0)$*, and* $\mu_{Poor}(4.0)$*.*

(ii) Compute the rule firing strengths α_1 *and* α_2*.*

(iii) Determine the crisp rule outputs $y_1 = \mu^{-1}_{High}(\alpha_1)$ *and* $y_2 = \mu^{-1}_{Low}(\alpha_2)$*.*

(iv) Calculate the final defuzzified crisp tip percentage $y^* = \dfrac{\alpha_1 y_1 + \alpha_2 y_2}{\alpha_1 + \alpha_2}$*.*

Example 7.10. *A chemical process utilizes saturated steam at pressure* $P_s = 68.95\ kPa$ *and saturation temperature* $T_s = 362.7\ K$ *to heat an incoming benzene stream. The benzene enters the exchanger at an initial inlet temperature* $T_1 = 17^\circ C$ *(*$290.15\ K$*), resulting in a temperature difference of* $T_s - T_1 = 72.55\ K$*. The specific heat capacity of liquid benzene is* $C_p = 1.7543\ kJ \cdot (kg \cdot K)^{-1}$*.*

The governing physical design equation for the heat exchanger size metric AU *(where* A *is heat transfer surface area and* U *is overall heat transfer coefficient) is expressed as:*

$$AU = wC_p \ln\left(\frac{T_s - T_1}{\Delta T_{app}}\right) \tag{7.44}$$

System Variables

- ***Input 1*** *(*$x_1 = w$*):* *Mass flow rate of benzene (kg/s).*
- ***Input 2*** *(*$x_2 = \Delta T_{app}$*):* *Temperature approach at the exchanger outlet (K).*
- ***Output*** *(*$y = AU$*):* *Heat exchanger size metric (kW/K).*

Fuzzy Rule Base

- ***Rule 1*** *(*R_1*):* ***IF*** w *is* A_1^1 *(large flow rate)* ***AND*** ΔT_{app} *is* A_2^1 *(small approach),* ***THEN*** AU *is* B^1 *(large exchanger).*
- ***Rule 2*** *(*R_2*):* ***IF*** w *is* A_1^2 *(small flow rate)* ***OR*** ΔT_{app} *is* A_2^2 *(large approach),* ***THEN*** AU *is* B^2 *(small exchanger).*
- ***Rule 3*** *(*R_3*):* ***IF*** w *is* A_1^2 *(small flow rate)* ***AND*** ΔT_{app} *is* A_2^1 *(small approach),* ***THEN*** AU *is* B^1 *(large exchanger).*

The fuzzy system is to be evaluated at the following specific Operating Point:

$$w = 1300\ kg/s, \qquad \Delta T_{app} = 6.5\ K \tag{7.45}$$

(i) Evaluate the crisp baseline value of AU using the physical governing equation at the given operating point.

(ii) Formulate the membership functions for the antecedent sets ($A_1^1, A_1^2, A_2^1, A_2^2$) and consequent sets ($B^1, B^2$).

(iii) Compute the rule firing strengths $\alpha_1, \alpha_2, \alpha_3$ under standard fuzzy T-norm ($\min$) and S-norm ($\max$) operators.

(iv) Determine the final defuzzified metric AU^* using Mamdani, Sugeno (TSK), and Tsukamoto inference methods.

7.4 Summary Comparison of System Outcomes

Table 11: Comparison of the three inference methods.

	Mamdani	**Sugeno (TSK)**	**Tsukamoto**
Consequent	fuzzy set (truncated or scaled)	crisp function $z = f(x, y)$	monotonic (shoulder) membership function
Rule output	fuzzy set	crisp value z_k	crisp value z_k
Overall output	aggregate then defuzzify (e.g. centroid)	weighted average	weighted average
Defuzzification	needed (time-consuming)	not needed	not needed
Typical use	expert knowledge, control	data-driven modelling, optimisation	specific situations only

A Appendix: Triangular Norms (T-Norms), S-Norms, and Fuzzy Operators

In fuzzy set theory, standard set operations (intersection, union, and complement) on crisp sets are generalized to continuous membership values in the unit interval $[0, 1]$ using binary operators known as **Triangular Norms (T-norms)** and **Triangular Conorms (S-norms)**.

A.1 1. Triangular Norms (T-Norms) — Fuzzy Intersection

A T-norm is a function $T : [0, 1] \times [0, 1] \to [0, 1]$ modeling the **fuzzy intersection (AND)** operation. For all $x, y, z \in [0, 1]$, a valid T-norm must satisfy four fundamental axioms:

1. **Boundary Condition:** $T(x, 1) = x$
2. **Monotonicity:** $y \leq z \implies T(x, y) \leq T(x, z)$
3. **Commutativity:** $T(x, y) = T(y, x)$
4. **Associativity:** $T(T(x, y), z) = T(x, T(y, z))$

Common T-Norm Operators

Operator Name	Mathematical Definition $T(a, b)$	Properties / Context
Minimum (Zadeh)	$\min(a, b)$	Standard, idempotent (min(
Algebraic Product	$a \cdot b$	Smooth, strictly monotonic
Bounded Difference (Lukasiewicz)	$\max(0, a + b - 1)$	Subadditive
Drastic Product	$\begin{cases} a, & b = 1 \\ b, & a = 1 \\ 0, & a, b < 1 \end{cases}$	Smallest possible T-norm (T

A.2 2. Triangular Conorms (S-Norms) — Fuzzy Union

An S-norm (or T-conorm) is a function $S : [0, 1] \times [0, 1] \to [0, 1]$ modeling the **fuzzy union (OR)** operation. S-norms are dual to T-norms under De Morgan's laws: $S(a, b) = 1 - T(1 - a, 1 - b)$.

The governing axioms for S-norms ($x, y, z \in [0, 1]$):

1. **Boundary Condition:** $S(x, 0) = x$
2. **Monotonicity:** $y \leq z \implies S(x, y) \leq S(x, z)$
3. **Commutativity:** $S(x, y) = S(y, x)$
4. **Associativity:** $S(S(x, y), z) = S(x, S(y, z))$

Common S-Norm Operators

A.3 3. Fuzzy Complements (N-Norms)

A fuzzy complement function $N : [0, 1] \to [0, 1]$ generalizes logical **NOT**.

Operator Name	**Mathematical Definition** $S(a,b)$	**Properties / Context**
Maximum (Zadeh)	$\max(a,b)$	Standard, idempotent ($\max(a,a) =$
Algebraic Sum	$a+b-a\cdot b$	Probabilistic sum
Bounded Sum (Lukasiewicz)	$\min(1,a+b)$	Superadditive
Drastic Sum	$\begin{cases} a, & b=0 \\ b, & a=0 \\ 1, & a,b>0 \end{cases}$	Largest possible S-norm ($S \le S_D$)

- **Axioms:** $N(0)=1$, $N(1)=0$, and if $a \le b$, then $N(a) \ge N(b)$ (monotonic decreasing).
- **Standard (Zadeh) Complement:** $N(a)=1-a$
- **Sugeno Class ($s>-1$):** $N_s(a)=\frac{1-a}{1+s\cdot a}$
- **Yager Class ($w>0$):** $N_w(a)=(1-a^w)^{1/w}$

A.4 4. Parametric Operator Families Aggregators

To provide tunable flexibility between standard intersection (min) and union (max), parameterized norms and averaging operators are used:

1. **Yager T-Norms and S-Norms ($p>0$):**

$$T_p(a,b)=1-\min\left(1,\left((1-a)^p+(1-b)^p\right)^{1/p}\right) \tag{A.1}$$

   As $p\to\infty$, $T_p(a,b)\to\min(a,b)$.

2. **Averaging Operators (Compensatory Aggregation):** Unlike T-norms (which are bounded above by min) and S-norms (bounded below by max), averaging operators satisfy:

$$\min(a,b)\le M(a,b)\le\max(a,b) \tag{A.2}$$

   Examples include the **Weighted Generalized Mean** $M_\alpha(a,b)=(w_1a^\alpha+w_2b^\alpha)^{1/\alpha}$ and **Ordered Weighted Averaging (OWA)** operators.

A.5 5. Summary Hierarchy of Norm Bounds

For any valid dual pair of T-norm and S-norm operators applied to $a,b\in[0,1]$:

$$T_D(a,b)\le\max(0,a+b-1)\le a\cdot b\le\min(a,b)\le S(a,b)\le\max(a,b)\le a+b-a\cdot b\le\min(1,a+b)\le S_D(a,b) \tag{A.3}$$